

MCS-221

Data Warehousing and Data Mining

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December 2021

Qus 1 :

(a) Define a Data Warehouse. List and explain the four characteristics of a Data Warehouse.

Ans :

Definition of Data Warehouse

A Data Warehouse (DW) is a centralized repository that stores large volumes of historical and current data from multiple heterogeneous sources (databases, files, ERP, CRM, etc.).

It is designed to support decision making, reporting, and analysis, rather than day-to-day transaction processing.

 **In simple words:**

A Data Warehouse is a subject-oriented, integrated, time-variant, and non-volatile collection of data that helps in decision support.

Four Characteristics of a Data Warehouse (Inmon's definition)

1. Subject-Oriented

- Data is organized around major subjects of the business (e.g., customers, sales, products, suppliers), rather than daily operations

- Helps managers analyze business performance in specific areas.
- Example: Instead of storing raw sales transactions, the warehouse stores aggregated information such as total sales by region, product, and time period.

2. Integrated

- Data from multiple heterogeneous sources (databases, files, applications) is cleaned, transformed, and unified before being loaded into the warehouse.
- Inconsistencies (e.g., different naming conventions, units, or formats) are resolved.
- Example: “Cust_ID” from CRM and “Customer_No” from ERP are standardized into a single “CustomerID” field.

3. Time-Variant

- Data is stored with a time dimension, allowing historical analysis.
- Unlike operational databases that focus on the present, the warehouse maintains snapshots over time.
- Example: A sales data warehouse can show product sales trends for the last 5 years, enabling forecasting.

4. Non-Volatile

- Once data enters the warehouse, it is not updated or deleted—only read and appended.
- Ensures stable and consistent data for analysis.
- Example: A sales transaction from 2020 remains unchanged in the warehouse even if the source operational system updates it.

Summary Table

Characteristic	Meaning	Example
Subject-Oriented	Organized by subject areas	Sales, Customers, Products
Integrated	Data from multiple sources, unified	CRM + ERP → One Customer ID

Time-Variant Historical data is maintained Sales over 5 years

Non-Volatile Data is read-only, not updated 2020 sales remain unchanged

“A subject-oriented, integrated, time-variant, and non-volatile collection of data that supports decision-making processes.”

(b) Explain the following OLAP architectures and draw their architectural diagram :

(i) Multidimensional Online Analytical Processing (MOLAP)

(ii) Hybrid Online Analytical Processing (HOLAP)

Ans :

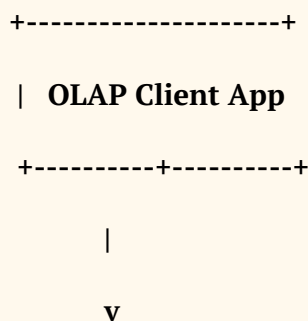
(i) Multidimensional OLAP (MOLAP)

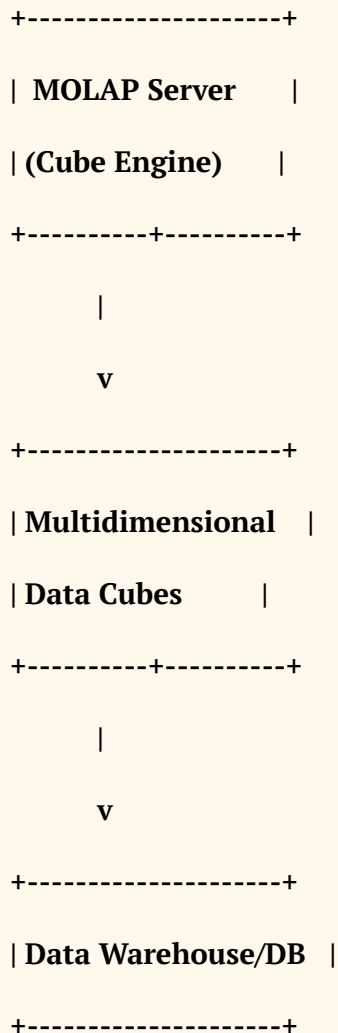
MOLAP is the traditional OLAP approach where data is stored in a multidimensional cube rather than relational databases.

Key Points:

- Data is pre-computed and stored in a multidimensional array (cube).
- Provides fast query performance (aggregation already done).
- Efficient for slicing, dicing, drilling down, roll-up operations.
- Drawback: Storage cost is high (cube takes more space).
- Suitable for small to medium datasets.

Architecture of MOLAP:





(ii) Hybrid OLAP (HOLAP)

HOLAP combines the advantages of MOLAP and ROLAP.

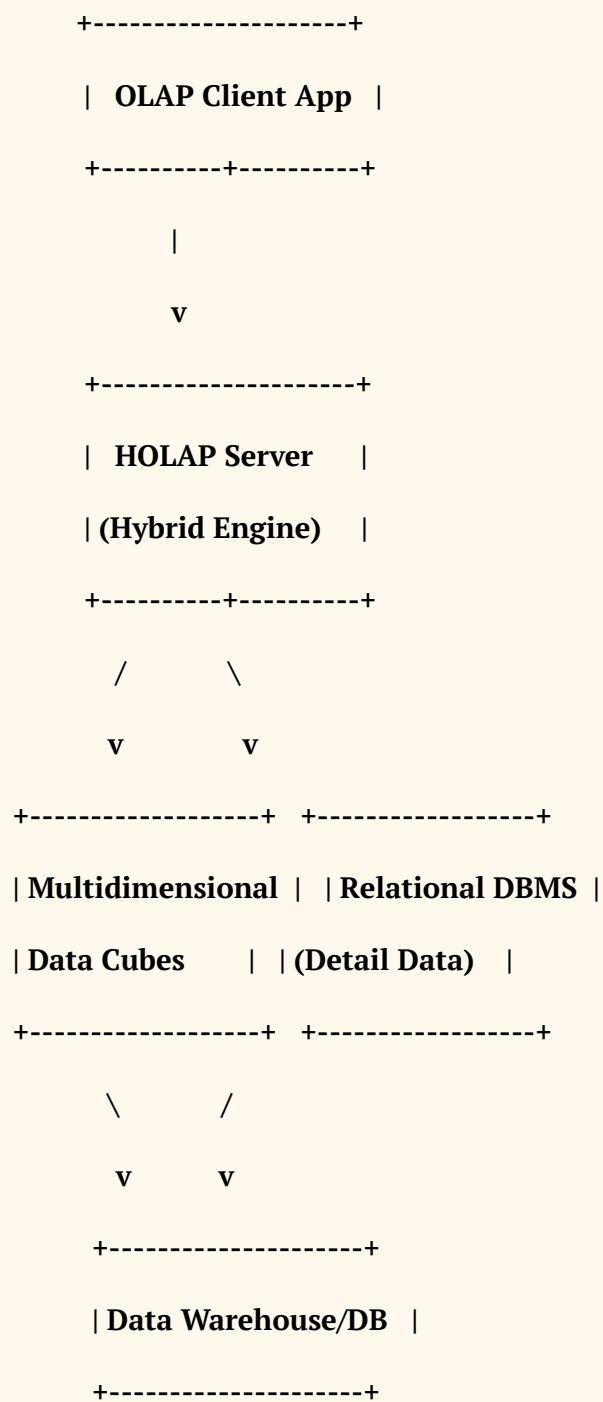
- ROLAP stores detail-level data in relational databases (good for scalability).
- MOLAP stores aggregated/summarized data in cubes (good for speed).
- HOLAP balances storage efficiency with performance.

Key Points:

- Aggregated data → stored in MOLAP cubes.
- Detailed data → stored in relational database (ROLAP).
- Provides fast queries on summaries and scalable detailed analysis.

- Suitable for large datasets.

Architecture of HOLAP:



✓ Summary:

- MOLAP = Fast queries, cube-based, storage-heavy.
- HOLAP = Hybrid approach, balances scalability (ROLAP) + performance (MOLAP).

(c) Define “data cleaning” which is a data preprocessing technique. In this context, explain the concept of Noisy data cleaning along with some suitable examples.

Ans :

Data Cleaning

Data Cleaning is a data preprocessing technique used in Data Warehousing and Data Mining. It refers to the process of identifying and correcting (or removing) errors and inconsistencies in data to improve data quality.

Main Goal:

- Handle missing values, duplicate records, incorrect entries, and noise.
- Ensure that the data is accurate, consistent, complete, and reliable before further analysis.

Noisy Data Cleaning

Noisy Data means data that has errors, outliers, or random variations that do not represent the true characteristics of information.

- Noise can occur due to faulty sensors, human errors in data entry, transmission errors, or measurement issues.
- Cleaning noisy data involves applying statistical and mathematical methods to smooth or filter the data.

Techniques for Noisy Data Cleaning

1. Binning

- Sort data and then apply smoothing techniques (like mean or median).

- **Example:**
 Raw data = {23, 25, 89, 24, 22, 900}
 Here, 900 is noisy (outlier).
 Using binning, the group average (say 30) can replace 900.

2. Regression

- **Fit a regression line/curve to the data and smooth out noise.**
- **Example:**
 If sales data has random spikes due to entry errors, a regression line will approximate the correct trend.

3. Clustering

- **Group similar data.**
- **Noise or outliers will appear in clusters with very few members.**
- **Example:**
 Customer ages = {25, 27, 26, 28, 120}.
 “120” is noise, since it doesn’t belong to the normal cluster.

4. Outlier Detection

- **Detect and remove extreme values.**
- **Example:**
 Income data = {20,000; 25,000; 30,000; 5,000,000}.
 The value 5,000,000 is noise (entry error).

Examples of Noisy Data Cleaning

- **Sensor Data:** A temperature sensor reading: 25°C, 26°C, 500°C, 27°C → 500°C is noise, removed by smoothing.
- **Survey Data:** Someone entering "999" for age instead of "29". → Corrected or discarded.
- **Stock Market Data:** Sudden false spike due to a data feed error.

✓ Summary for Exam:

- Data Cleaning improves data quality by correcting errors.
- Noisy Data Cleaning deals with removing random errors, outliers, and inconsistencies using techniques like binning, regression, clustering, and outlier detection.

(d) What is a Decision Tree ? How is it useful in classification ? With the help of an example, explain the process of construction of a decision tree and its representation.

Ans :

Decision Tree

Definition:

A Decision Tree is a tree-structured model used for classification and prediction.

- Each internal node represents a test on an attribute (e.g., "Is Age < 30?").
- Each branch represents the outcome of the test (e.g., "Yes" or "No").
- Each leaf node represents a class label (decision).

Usefulness in Classification

- Breaks a complex decision-making process into a simple series of tests.
- Easy to understand, interpret, and visualize.
- Can handle both categorical and numerical data.
- Used in customer segmentation, medical diagnosis, fraud detection, loan approval, etc.

Steps in Constructing a Decision Tree

1. Select the best attribute (feature) to split data (using measures like *Information Gain*, *Gini Index*).
2. Create a decision node based on that attribute.
3. Split the dataset into subsets based on attribute values.

4. Repeat recursively for each subset until:

- All records in a subset belong to the same class, OR
- No more attributes remain.

Example of Decision Tree Construction

Example Problem:

We want to predict whether a person will play tennis based on weather conditions.

Training Dataset:

Outlook	Temperature	Humidity	Wind	Play Tennis
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes

Rain	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rain	Mild	High	Strong	No

Step 1: Choose Root Node

- Use Information Gain.
- The attribute Outlook gives the highest information gain.
- So, root node = Outlook.

Step 2: Split on Outlook

- Outlook = Sunny → depends on Humidity.
- Outlook = Overcast → Always Yes.
- Outlook = Rain → depends on Wind.

Final Decision Tree Representation

```

[Outlook]
/  |  \
Sunny Overcast Rain

```

/			\	
[Humidity]		Yes	[Wind]	
/ \			/ \	
High	Normal		Weak	Strong
No	Yes		Yes	No

Interpretation:

- If Outlook = Overcast → Play = Yes.
- If Outlook = Sunny AND Humidity = High → Play = No.
- If Outlook = Rain AND Wind = Weak → Play = Yes.

✓ Summary for Exam:

- A Decision Tree is a classification model where nodes = tests, branches = outcomes, leaves = decisions.
- Constructed using attribute selection (Info Gain, Gini).
- Example: "Play Tennis" dataset, with root = Outlook, splits = Humidity/Wind.

Qus 2.

(a) What is Text Mining ? Where is it used ? Explain any two text mining techniques with the help of a suitable example for each.

Ans :

Text Mining

Definition:

Text Mining (or Text Data Mining) is the process of extracting useful information, patterns, and knowledge from large amounts of unstructured textual data using techniques from data mining, natural language processing (NLP), and machine learning.

Where is it used?

- **Sentiment Analysis:** Understanding customer opinions on social media, reviews.
- **Spam Filtering:** Identifying spam vs. non-spam emails.
- **Document Classification:** Categorizing research papers, news, or legal documents.
- **Chatbots/Virtual Assistants:** Understanding user queries.
- **Fraud Detection & Healthcare:** Mining medical records, detecting suspicious claims.

Two Text Mining Techniques

1. Text Classification

Definition:

Automatically assigning a text document into one or more predefined categories.

Method:

- Uses algorithms like Naïve Bayes, Decision Trees, or Support Vector Machines (SVMs).

Example:

- **Email classification:**
 - Input: "Win ₦10,000 now! Click here!"
 - Classifier learns from past labeled emails.
 - Output: Classified as Spam.

2. Sentiment Analysis

Definition:

Analyzing text to determine whether the sentiment is positive, negative, or neutral.

Method:

- Uses lexicon-based approaches (dictionary of positive/negative words) or machine learning models.

Example:

- Customer Review: *“The camera quality of this phone is excellent but the battery is poor.”*
 - Sentiment analysis identifies:
 - Positive aspect → “camera quality: excellent”
 - Negative aspect → “battery: poor”
 - Output: Mixed Sentiment.

✓ Summary for Exam:

- Text Mining = extracting knowledge from unstructured text.
- Applications = spam detection, sentiment analysis, document classification, etc.
- Techniques:
 1. Text Classification → classifies text into predefined categories (e.g., spam filter).
 2. Sentiment Analysis → detects emotions/attitudes in text (e.g., customer reviews).

(b) In the context of mining multimedia data on the web, explain the following terms :

- Page Rank
- Hits
- Page Layout Analysis
- Vision Page Segmentation

Ans : Mining Multimedia Data on the Web – Key Terms

(i) PageRank

- **Definition:** PageRank is an algorithm developed by Google to rank web pages in search engine results.
- **Concept:**
 - A page is considered important if many other important pages link to it.
 - Uses the idea of a “random surfer” who randomly clicks on links.
- **Formula (simplified):**

$$PR(A) = (1-d) + d \sum \frac{PR(T_i)}{C(T_i)}$$
 where:
 - $PR(A)$ = PageRank of page A
 - d = damping factor (usually 0.85)
 - T_i = pages linking to A
 - $C(T_i)$ = number of outbound links from page T_i

✓ Example:

If many popular tech blogs link to your portfolio site, your site's PageRank increases.

(ii) HITS (Hyperlink-Induced Topic Search)

- **Definition:** An algorithm proposed by Jon Kleinberg, used for link analysis of web pages.
- **Concept:** Each page has two scores:
 1. Hub Score – if the page points to many good authorities.
 2. Authority Score – if the page is linked by many good hubs.
- **Working:** Iterative updates where hubs and authorities reinforce each other.

✓ Example:

- A "list of best colleges" page acts as a hub (links to many universities).
- Each university's official site becomes an authority.

(iii) Page Layout Analysis

- **Definition:** The process of analyzing the structure and arrangement of content (text, images, tables, headings) on a web page or document.

- **Purpose:** Helps in content extraction, indexing, and readability improvement.
- **Steps:**
 - Detect headers, footers, paragraphs, tables, images.
 - Understand logical reading order.

✓ **Example:**

In an online newspaper, page layout analysis separates news article body, advertisements, and sidebars for better indexing.

(iv) Vision-based Page Segmentation (VIPS)

- **Definition:** A technique that mimics human visual perception to divide a web page into semantically meaningful segments.
- **Approach:**
 - Uses visual cues (fonts, spacing, borders, colors) to segment a page.
 - Unlike plain HTML tag-based segmentation, it focuses on how users perceive the page visually.
- **Use:** Improves search engines, information extraction, and web content mining.

✓ **Example:**

On an e-commerce site:

- Product image + price + “Add to Cart” button = one segment.
- Recommendations section = another segment.

✓ **Summary Table for Revision**

Term	Meaning	Example
PageRank	Ranks pages by importance using inbound links	Tech blogs linking to your site increase its rank
HITS	Hub & Authority model for link analysis	“Best colleges list” = Hub, “College website” = Authority

Page Layout Analysis	Identifies structure (header, body, images)	Newspaper: separates article vs. ads
Vision Page Segmentation	Splits web page by visual cues	E-commerce: product block vs. recommendations

Qus 3.

(a) Define a Data Lake. Explain the step-by-step process of creating a data lake.

Ans: Definition:

A Data Lake is a centralized storage repository that allows you to store structured, semi-structured, and unstructured data at any scale in its raw format.

- Unlike a Data Warehouse (which stores processed, structured data), a Data Lake stores raw data and processes it only when needed.
- Technologies: Hadoop, AWS S3, Azure Data Lake, etc

Step-by-Step Process of Creating a Data Lake

1. Data Ingestion

- Collect data from multiple sources → databases, IoT devices, logs, social media, images, etc.
- Example tools: Apache Kafka, AWS Kinesis.

2. Data Storage

- Store raw data (structured, semi-structured, unstructured) in a scalable storage system.
- Example: Hadoop HDFS, AWS S3, Azure Data Lake Storage.

3. Data Organization & Cataloging

- Use metadata to describe data (what it is, source, format).
- Helps in search, governance, and avoiding “data swamp”.
- Example: AWS Glue, Apache Hive Metastore.

4. Data Processing & Transformation

- Clean, filter, and process raw data when needed (ELT approach).
- Tools: Apache Spark, Databricks, AWS EMR.
- 5. Data Security & Governance
 - Apply access control, encryption, and auditing.
 - Ensures compliance (GDPR, HIPAA).
- 6. Data Consumption
 - Analysts, Data Scientists, and BI tools use the lake for analytics, machine learning, dashboards.
 - Example: Power BI, Tableau, Jupyter Notebooks.

✓ In short:

Data flows into the Lake (raw) → Catalogued → Processed when needed → Used by BI/ML applications.

(b) With the help of an example, explain the star schema dimensional model

Ans :

Ans : Definition:

The Star Schema is the most common dimensional data modeling technique used in data warehouses.

- A central fact table (quantitative data) connects to multiple dimension tables (descriptive attributes).
- Called a star because the schema looks like a star shape.

Components of Star Schema

1. Fact Table:
 - Stores measurable, quantitative data (facts).
 - Contains foreign keys referencing dimension tables.
 - Example: Sales Amount, Quantity Sold.
2. Dimension Tables:

- Stores descriptive attributes that describe facts.
- Example: Customer, Product, Time, Store.

Example: Retail Sales Data Warehouse

Fact Table: **Sales_Fact**

| Sale_ID | Date_Key | Product_Key | Customer_Key | Store_Key | Sales_Amount | Quantity |

Dimension Table: **Time_Dim**

| Date_Key | Date | Month | Quarter | Year |

Dimension Table: **Product_Dim**

| Product_Key | Product_Name | Category | Brand | Price |

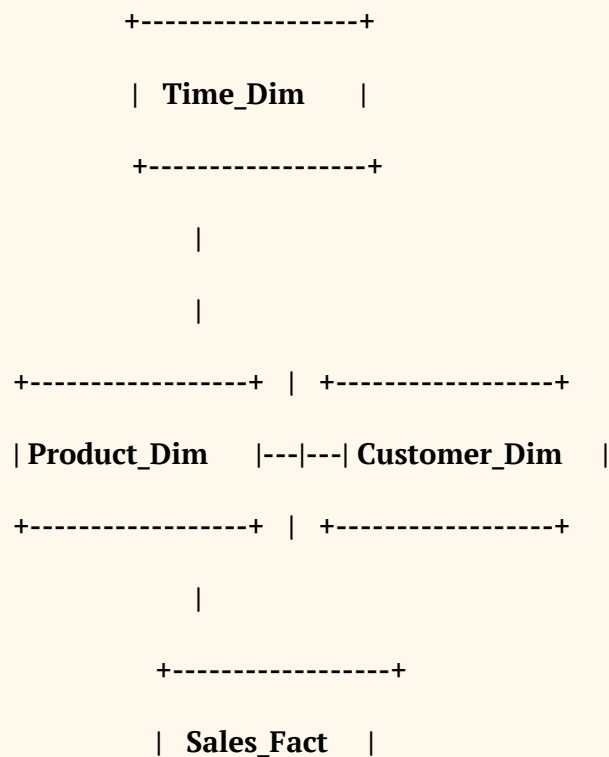
Dimension Table: **Customer_Dim**

| Customer_Key | Name | Gender | City | Age |

Dimension Table: **Store_Dim**

| Store_Key | Store_Name | Location | Manager |

Star Schema Representation (Diagram)



```

+-----+
|
|
+-----+
| Store_Dim |
+-----+

```

How it Works:

- A query like “*Find total sales of Electronics in 2024 by store*” →
 - Uses Sales_Fact (Sales_Amount, Quantity)
 - Joins with Product_Dim (Category = Electronics)
 - Joins with Time_Dim (Year = 2024)
 - Joins with Store_Dim (Store wise).

✓ Summary for Exam:

- Data Lake: Raw data storage → Ingestion → Storage → Catalog → Processing → Security → Consumption.
- Star Schema: Central Fact Table + surrounding Dimension Tables → Simple, fast for queries.

Qus 4.

(a) What is Metadata ? What are its contents ? Justify how metadata can be an important component in Data Warehousing ? Also mention its types.

Ans :

Metadata in Data Warehousing

Definition:

- **Metadata is “data about data”.**
- **In the context of a data warehouse, metadata describes:**
 - What the data is,
 - Where it comes from,
 - How it is transformed, and
 - How it is structured and used.

Contents of Metadata

Metadata in a data warehouse typically contains:

- 1. Structural Information**
 - Table names, columns, datatypes, indexes, keys.
 - Example: Customer_ID (Integer, Primary Key).
- 2. Data Mapping and Transformation Rules**
 - Source-to-target mappings.
 - Transformation logic applied during ETL (Extract–Transform–Load).
- 3. Business Definitions**
 - Meaning of attributes and measures.
 - Example: “Revenue = Total Sales – Discounts”.
- 4. Data Lineage (Provenance)**
 - Where the data came from (source systems).
 - How it has changed over time.
- 5. Operational Information**
 - Refresh frequency, update time, load history.
- 6. Access and Security Information**
 - Who can access the data, roles, permissions.

Importance of Metadata in Data Warehousing

Metadata is a key component because:

- 1. Acts as a Roadmap – Helps users locate and understand data.**
- 2. Supports ETL – Guides extraction, transformation, and loading of data.**
- 3. Improves Query Performance – Optimizers use metadata (indexes, partitions).**

4. Ensures Consistency – Common business definitions across the enterprise.
5. Data Governance & Security – Tracks data ownership, access control.
6. Self-Service BI – Business users can explore data without needing IT intervention.

👉 Without metadata, a data warehouse is like a library without a catalog.

Types of Metadata in Data Warehousing

1. Technical Metadata
 - Details about structure and storage of data.
 - Includes schemas, datatypes, indexes, ETL mappings.
2. Business Metadata
 - Describes the meaning and context of data in business terms.
 - Example: Customer, Sales, Profit definitions.
3. Operational Metadata
 - Information about warehouse operations.
 - Includes load schedules, refresh rates, error logs, performance stats.

✅ Summary (Exam-Ready)

- Metadata = "data about data".
- Contents = structure, mappings, definitions, lineage, security, operations.
- Importance = roadmap for users, supports ETL, improves performance, ensures governance.
- Types = Technical, Business, Operational.

(b) Explain the three components of ETL. Also mention how to improve the ETL performance.

Ans :

ETL (Extract – Transform – Load)

ETL is the core process of Data Warehousing that moves data from multiple heterogeneous sources to the data warehouse in a clean and usable form.

Three Components of ETL

1. Extraction

- **Definition:** Extracting data from multiple source systems (databases, flat files, ERP, IoT, social media, etc.).
- **Tasks:**
 - Identify source systems.
 - Extract full or incremental data.
 - Handle structured, semi-structured, and unstructured formats.
- **Example:** Pulling customer data from Oracle DB + sales logs from CSV files.

2. Transformation

- **Definition:** Converting extracted data into a clean, consistent, and meaningful format for analysis.
- **Tasks:**
 - Data cleaning (remove duplicates, handle missing values).
 - Data integration (merging data from multiple sources).
 - Data conversion (date formats, currency, units).
 - Business rule application (calculating profit = sales – cost).
- **Example:** Converting all dates into **YYYY-MM-DD**, currency into INR, correcting null values.

3. Loading

- **Definition:** Loading the transformed data into the Data Warehouse or Data Lake.
- **Types of Loading:**
 - Full Load → Entire dataset loaded (first-time load).
 - Incremental Load → Only changes (inserts/updates/deletes) since last load.
- **Example:** Loading daily sales summary into the **Sales_Fact** table of a warehouse.

Improving ETL Performance

To make ETL faster and more efficient:

1. **Efficient Extraction**
 - Use incremental extraction instead of full extraction.
 - Apply filters to fetch only relevant data.
2. **Parallel Processing**
 - Run ETL jobs in parallel (multi-threading, partitioning).
3. **Push-down Transformation**
 - Perform transformations at the database/source level instead of ETL tool, reducing data movement.
4. **Use of Staging Area**
 - Extract raw data into a staging area before transforming, to reduce load on the warehouse.
5. **Indexing and Partitioning**
 - Index fact tables and partition large datasets to speed up loading and querying.
6. **Efficient Data Cleaning**
 - Use bulk operations instead of row-by-row cleaning.
7. **Monitoring and Scheduling**
 - Automate ETL workflows and run during off-peak hours.

✓ Summary (Exam-Ready)

- **ETL Components:**
 1. Extraction – collect raw data from sources.
 2. Transformation – clean, integrate, convert, and apply business rules.
 3. Loading – move transformed data into warehouse.
- **Performance Improvements:** Use incremental loads, parallel processing, push-down transformations, staging area, indexing/partitioning, and automated scheduling.

Qus 5. Write short notes on any four of the following :

(a) Data Marts

(b) Association Rule Generation

(c) Apriori Algorithm

(d) Clustering and its Methods

(e) Bayes' Theorem

Ans:

(a) Data Marts

- **Definition:**
A Data Mart is a subset of a Data Warehouse, focused on a specific business line, department, or subject area (e.g., Sales, Marketing, Finance).
- **Types:**
 - Dependent Data Mart – Built from an existing data warehouse.
 - Independent Data Mart – Built directly from operational systems.
- **Advantages:**
 - Faster access for specific business needs.
 - Cost-effective compared to a full data warehouse.
- **Example:**
A Sales Data Mart may store only sales-related facts (sales amount, region, product).

(b) Association Rule Generation

- **Definition:**
Association rule generation is a data mining technique used to find relationships (co-occurrences) among items in large datasets.
- **Form:**
If $X \rightarrow Y$, then customers who buy X are likely to buy Y.
- **Measures:**
 - **Support:** Probability that X and Y occur together.
 - **Confidence:** Probability that Y occurs given X.
 - **Lift:** Strength of the rule compared to random chance.
- **Example:**
Rule: {Milk} $\rightarrow$ {Bread}
 - Support = 30% (in 30% of transactions both appear).
 - Confidence = 70% (70% of milk buyers also buy bread).

(c) Apriori Algorithm

- **Definition:**
Apriori is a classic algorithm used for frequent itemset mining and association rule learning.
- **Working Steps:**
 - Identify frequent 1-itemsets (items with minimum support).
 - Generate k-itemsets from (k-1)-itemsets using the Apriori property:
“If an itemset is frequent, all its subsets are also frequent.”
 - Prune infrequent itemsets.
 - Generate association rules using confidence.
- **Example:**
Transactions: {Milk, Bread, Butter}, {Milk, Bread}, {Bread, Butter}
 - Frequent Itemsets: {Milk, Bread}, {Bread, Butter}
 - Rule: Milk $\rightarrow$ Bread (Confidence = 80%).

(d) Clustering and its Methods

- **Definition:**
Clustering is an unsupervised learning technique that groups data objects into clusters such that:
 1. Objects in the same cluster are similar,
 2. Objects in different clusters are dissimilar.
- **Applications:** Customer segmentation, Image processing, Document grouping.
- **Methods:**
 1. Partitioning Methods (e.g., K-Means) – Divides data into k clusters.
 2. Hierarchical Methods – Creates a tree (dendrogram) of clusters.
 3. Density-based Methods (e.g., DBSCAN) – Forms clusters based on data density.
 4. Grid-based Methods – Uses multi-dimensional grids (e.g., STING).
- **Example:** Grouping customers into clusters based on age & income.

(e) Bayes' Theorem

- **Definition:**

Bayes' Theorem is a probabilistic technique used for classification in data mining and machine learning.

- **Formula:**

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

- $P(A|B)$: Posterior probability (probability of A given B).
- $P(A)$: Prior probability of A.
- $P(B|A)$: Likelihood (probability of B given A).
- $P(B)$: Evidence.

- **Example:**

Spam filtering:

- A = Email is Spam
- B = Email contains word "Free"
- Calculate $P(\text{Spam} | \text{Free})$ to decide classification.

 Summary for Exam:

- Data Mart – subset of DW, subject-oriented (e.g., Sales Data Mart).
- Association Rule – finds item relationships ($X \rightarrow Y$).
- Apriori Algorithm – generates frequent itemsets using Apriori property.
- Clustering – unsupervised grouping; methods = K-Means, Hierarchical, DBSCAN.
- Bayes' Theorem – probability-based classification (Naïve Bayes).

June 2022

Q.1 (a) What is Association Rule Mining? With reference to this, explain the concepts given below:

- (i) Support Count
- (ii) Frequent Itemset
- (iii) Association Rule
- (iv) Rule Evaluation Metrics

Ans:

Association Rule Mining

- **Definition:** Association Rule Mining is a data mining technique used to discover hidden relationships and patterns among items in large datasets.
- **Goal:** Find rules of the form $X \rightarrow Y$, meaning *if X occurs, then Y is likely to occur*.
- **Applications:** Market basket analysis, recommendation systems, fraud detection.

(i) Support Count

- **Definition:** Number of transactions in which a particular itemset appears.
- **Formula:**

$$\text{Support_count}(X) = |\{t \in T \mid X \subseteq t\}|$$

$$\text{Support_count}(X) = |\{t \in T \mid X \subseteq t\}|$$
 where T = total transactions.
- **Example:**
 In 10 transactions, {Milk, Bread} appears in 3 → Support Count = 3.

(ii) Frequent Itemset

- **Definition:** An itemset whose support count $\geq$ minimum support threshold.
- **Example:**
 If min support = 30%, and {Milk, Bread} appears in 40% of transactions → it is a frequent itemset.

(iii) Association Rule

- **Definition:** An implication expression of the form $X \rightarrow Y$, where X and Y are itemsets and $X \cap Y = \emptyset$.
- **Meaning:** If X occurs, then Y is likely to occur.
- **Example:** $\{\text{Milk}\} \rightarrow \{\text{Bread}\} \rightarrow$ Customers who buy milk also buy bread.

(iv) Rule Evaluation Metrics

Rules are evaluated using:

1. Support:

$\text{Support}(X \rightarrow Y) = \frac{\text{Transactions}(X \cup Y)}{\text{Total Transactions}}$
 $\rightarrow$ Probability both X and Y occur together.

2. Confidence:

$\text{Confidence}(X \rightarrow Y) = \frac{\text{Support}(X \cup Y)}{\text{Support}(X)}$
 $\rightarrow$ Probability Y occurs given X occurs.

3. Lift:

$\text{Lift}(X \rightarrow Y) = \frac{\text{Confidence}(X \rightarrow Y)}{\text{Support}(Y)}$
 $\rightarrow$ Strength of the rule compared to random chance.

Example:

- 100 transactions.
- $\{\text{Milk}\} \rightarrow \{\text{Bread}\}$ appears in 30 transactions.
- $\text{Support} = 30/100 = 30\%$.
- If Milk appears in 40 transactions $\rightarrow \text{Confidence} = 30/40 = 75\%$.
- If Bread appears in 50 transactions $\rightarrow \text{Lift} = 0.75 / 0.50 = 1.5$ (>1 means strong rule).

✓ Summary for Exam:

- **Support Count** = raw frequency of itemset.
- **Frequent Itemset** = appears $\geq$ min support.

- Association Rule = $X \rightarrow Y$ relation.
- Metrics = Support, Confidence, Lift (measure strength of rules).

(b) Define classification. With the help of an example for each, explain the following classification models :

(i) Descriptive modeling

(ii) Predictive modeling

Ans:

Classification – Definition

- Classification is a supervised learning technique in data mining and machine learning.
- It assigns data items to one of the predefined categories (classes) based on input attributes.
- Uses a training dataset with known class labels to build a model, which is then used for classifying new data.

👉 **Examples:** Email classification (Spam / Not Spam), Medical diagnosis (Disease / No Disease), Credit approval (Approve / Reject).

(i) Descriptive Modeling

- **Definition:**
Descriptive modeling describes the characteristics of data classes and provides insight into relationships among attributes.
 - It focuses on understanding the structure of existing data.
 - Often represented as rules, decision trees, or clusters.
- **Example:**
A bank uses descriptive modeling to analyze loan default patterns.
 - Rule discovered: “80% of defaulters have income < ₦30,000 and credit score < 600”.

- This does not predict future defaulters directly, but helps understand why defaults happen.

(ii) Predictive Modeling

- **Definition:**
Predictive modeling builds a model from historical data and uses it to predict unknown or future outcomes.
 - Focus is on forecasting and classification of unseen data.
 - Methods: Decision Trees, Naïve Bayes, Neural Networks, SVMs.
- **Example:**
An e-commerce company builds a predictive model using past purchase history to predict whether a new customer will buy a product.
 - If attributes: *Age = 25, Browsing time = High, Location = Metro City* → Model predicts “Will Buy”.

✓ Summary (Exam-Ready)

- **Classification:** Assigns objects to predefined classes.
- **Descriptive Modeling:** Explains existing patterns in data (Example: Loan default characteristics).
- **Predictive Modeling:** Forecasts class labels for unseen data (Example: Predicting whether a customer will buy).

(c) Describe the snowflake schema multidimensional modeling technique. Give an example and illustrate it. List its pros and cons.

Ans :

Definition:

- A Snowflake Schema is a type of multidimensional schema used in data warehouses.
- It is a normalized extension of the Star Schema, where:
 - Dimension tables are split into multiple related tables, forming a structure like a *snowflake*.

- Normalization reduces redundancy.

👉 **Key Idea:** Dimensions are organized into hierarchies and broken into multiple related tables.

Example: Retail Sales Data Warehouse

Fact Table: Sales_Fact

| Sale_ID | Date_Key | Product_Key | Customer_Key | Store_Key | Sales_Amount | Quantity |

Dimension Tables (Normalized):

1. Product Dimension

- Product_Dim(Product_Key, Product_Name, Category_Key)
- Category_Dim(Category_Key, Category_Name, Department_Key)
- Department_Dim(Department_Key, Department_Name)

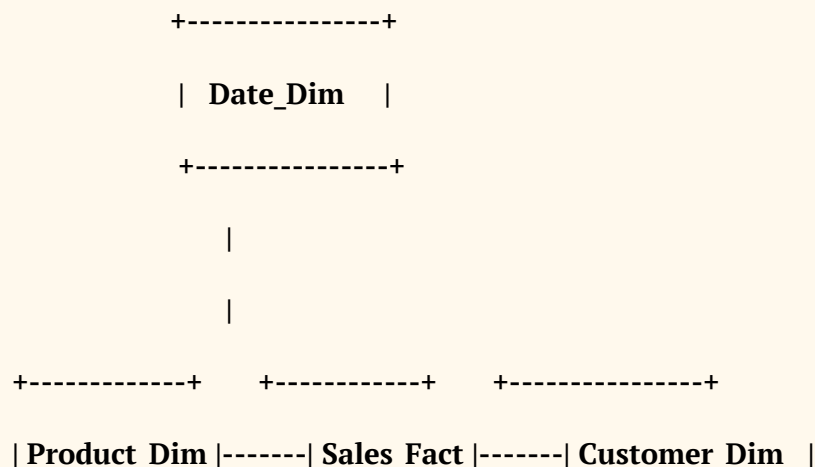
2. Customer Dimension

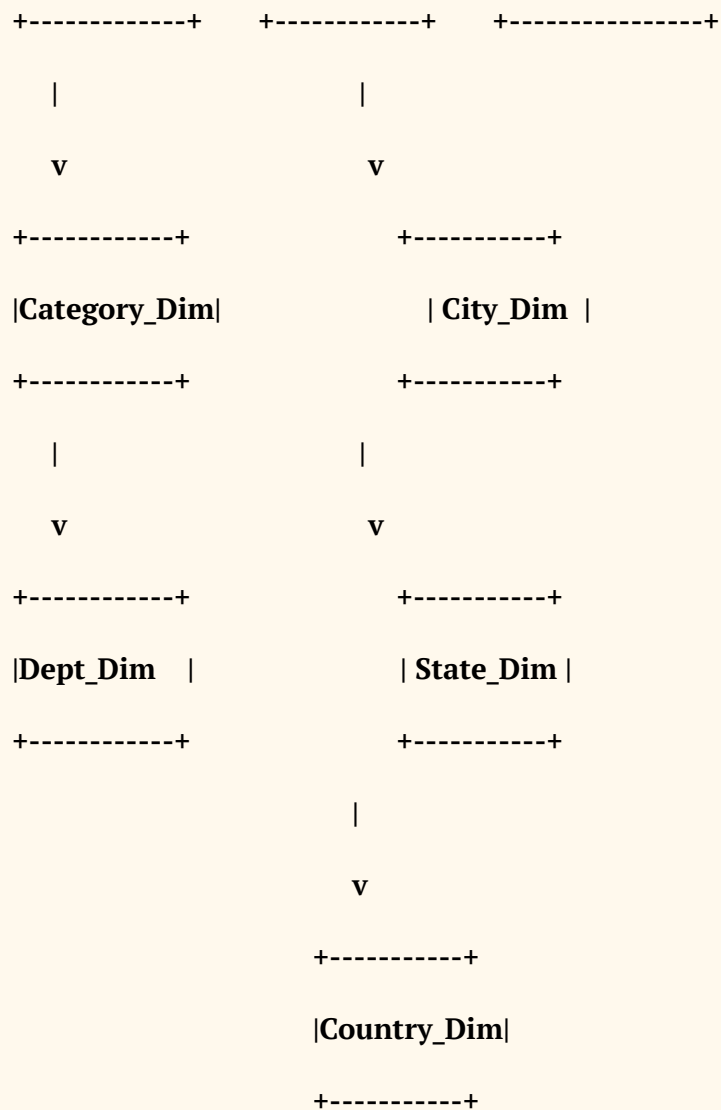
- Customer_Dim(Customer_Key, Name, City_Key)
- City_Dim(City_Key, City_Name, State_Key)
- State_Dim(State_Key, State_Name, Country_Key)
- Country_Dim(Country_Key, Country_Name)

3. Time Dimension

- Time_Dim(Date_Key, Day, Month_Key, Year)
- Month_Dim(Month_Key, Month_Name, Quarter)

Snowflake Schema Diagram





Pros of Snowflake Schema

1. **Normalized Structure** → Removes redundancy, saves storage space.
2. **Better Data Integrity** → Updates are consistent (less duplication).
3. **Supports Hierarchies** → Useful for drill-down (e.g., Product → Category → Department).

Cons of Snowflake Schema

1. **Complex Queries** → Requires multiple joins → slower query performance compared to Star Schema.
2. **Difficult for Users** → Harder for business users to understand.
3. **Longer Query Execution Time** → Joins increase computational cost.

✓ Summary (Exam-Ready)

- **Snowflake Schema** = Normalized Star Schema with split dimension tables.
- **Example** = Retail Sales: Product → Category → Department hierarchy.
- **Pros** = Less redundancy, better integrity, supports hierarchies.
- **Cons** = More complex queries, slower joins, harder for users.

(d) What is complex data modeling ? Where is this used ? Explain “Anchor” complex data model.

Ans :

Definition:

- **Complex Data Modeling** refers to advanced modeling techniques used when traditional Star Schema or Snowflake Schema are not sufficient to represent the data requirements.
- It is used for handling large-scale, highly normalized, historical, and flexible data structures in a Data Warehouse.
- It often focuses on scalability, agility, and flexibility to adapt to changing business requirements.

Where is it used?

- In large enterprises where data structures evolve continuously.
- When business rules change frequently (e.g., banking, telecom, insurance).
- In historical data warehouses that require versioning and auditing of data.
- In Big Data environments where complex relationships must be tracked.

Anchor Modeling (A Complex Data Model)

Definition:

- Anchor Modeling is a complex, agile, and temporal data modeling technique used in data warehouses.
- It is highly normalized (6NF) and focuses on flexibility and historization (tracking changes over time).
- Developed to support evolving databases with minimal impact on existing structures.

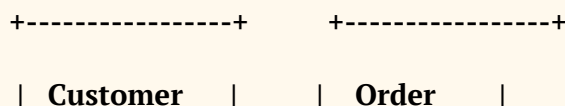
Key Concepts of Anchor Modeling

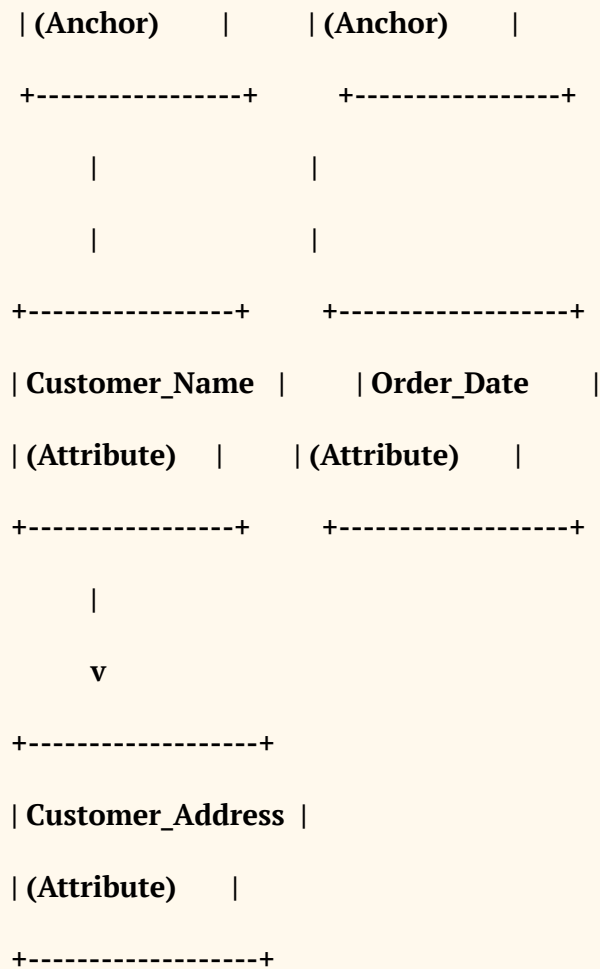
1. Anchor (Entity):
 - Represents a core business concept (e.g., Customer, Product).
 - Each anchor has a unique surrogate key.
2. Attribute:
 - Describes the anchor (e.g., Customer Name, Product Price).
 - Attributes are stored in separate tables, allowing historization.
3. Tie (Relationship):
 - Defines relationships between anchors.
 - Example: Customer ↔ Order.
4. Historization:
 - Changes in attributes are stored with validity time ranges → supports temporal queries.

Example: Customer Database

- Anchor: **Customer** (Customer_ID).
- Attribute:
 - Customer_Name (with timestamp → stores history when name changes).
 - Customer_Address (with timestamp → records address changes).
- Tie: Customer ↔ Order relationship stored separately.

Illustration (Simplified Diagram)





(Each attribute table keeps history with timestamps → e.g., address changes over time are preserved.)

Advantages of Anchor Modeling

- **Flexibility:** Easy to add new attributes without redesigning schema.
- **Temporal Support:** Naturally supports history tracking.
- **Scalability:** Suitable for large and evolving data warehouses.

Disadvantages

- **Highly Normalized:** Queries require many joins → can reduce performance.

- Complexity: More difficult for end-users to understand compared to Star Schema.

✓ Summary (Exam-Ready)

- Complex Data Modeling = advanced modeling for large, evolving data.
- Used in: Banking, Telecom, Insurance, Big Data warehouses.
- Anchor Modeling = a temporal, 6NF, flexible model with Anchors (entities), Attributes, Ties (relationships).
- Pros: Flexibility, temporal historization, scalable.
- Cons: Many joins, query complexity.

2. (a) List and describe all the four types of OLAP operations.

Ans :

OLAP Operations

In OLAP (Online Analytical Processing), different operations are used to analyze multidimensional data in a Data Warehouse.

The four basic OLAP operations are:

1. Roll-up (Aggregation)

- Definition:
The roll-up operation summarizes data by climbing up the hierarchy or by reducing dimensions.
- How it works:
 - Moving from detailed level → summarized level.
 - Example hierarchies: *Day* → *Month* → *Quarter* → *Year*.
- Example:
Sales data by Day → Rolled up to Month → Rolled up to Year.
- Use: Get a high-level overview of data.

2. Drill-down (Reverse of Roll-up)

- **Definition:**
Drill-down provides a more detailed view of data by moving down the hierarchy.
- **How it works:**
 - Moving from summary level → detailed level.
 - Example hierarchies: *Year* → *Quarter* → *Month* → *Day*.
- **Example:**
Sales data shown at Year level → Drill down to Month level → Further to Day level.
- **Use:** Perform detailed analysis of data.

3. Slice

- **Definition:**
Slice selects a single dimension value and produces a sub-cube by fixing one dimension.
- **How it works:**
 - Choose one value for one dimension, keep all other dimensions.
- **Example:**
Sales data with dimensions (*Product, Region, Time*) →
Slice on Time = 2024 → Cube shows only sales for year 2024.
- **Use:** Analyze a single perspective of data.

4. Dice

- **Definition:**
Dice selects a sub-cube by applying a range of values (conditions) on multiple dimensions.
- **How it works:**
 - Apply conditions on two or more dimensions.
- **Example:**
Sales data with dimensions (*Product, Region, Time*) →
Dice on (*Product* = “*Mobile, Laptop*” AND *Region* = “*West, North*” AND *Time* = *2023-2024*).
- **Use:** Focus on a specific subset of the data

OLAP Operation	Description	Example
Roll-up	Summarize data by climbing up hierarchy	Day → Month → Year
Drill-down	Go into detail by descending hierarchy	Year → Month → Day
Slice	Select single value on one dimension	Time = 2024
Dice	Select sub-cube using multiple conditions	Product = Mobile, Region = West

👉 Would you like me to also draw a 3D OLAP Cube diagram with Roll-up, Drill-down, Slice, and Dice marked on it? (Very scoring in exams if you draw).

(b) Define text mining. Mention any three applications of it. What are the various techniques used to analyze the web-usage patterns ?

Ans:

1. Text Mining – Definition

- Text Mining (also called Text Data Mining or Text Analytics) is the process of extracting useful patterns, knowledge, and insights from large amounts of unstructured text data using techniques from data mining, natural language processing (NLP), and machine learning.

👉 In short: It converts unstructured text into structured knowledge for decision-making.

2. Applications of Text Mining (Any Three)

1. Sentiment Analysis

- Analyzing reviews, social media, or survey responses to determine customer opinions (positive, negative, neutral).
- Example: Analyzing tweets about a new movie.

2. Spam Filtering

- Email classification into Spam or Not Spam using keyword frequency and ML models.

3. Document Classification & Clustering

- Grouping research papers, news articles, or legal documents into categories.
- Example: Classifying news into categories like Politics, Sports, Technology.

4. (Optional extra if needed) Chatbots & Virtual Assistants

- Understanding user queries and responding appropriately using mined text patterns.

3. Techniques to Analyze Web-Usage Patterns

Web Usage Mining = Discovering useful patterns from web logs (user clicks, navigation paths, session history).

The main techniques are:

1. Association Rule Mining

- Finds relationships among web pages that are frequently visited together.
- Example: *“Users who visit the Home Page also visit the Product Page 80% of the time.”*

2. Sequential Pattern Mining

- Identifies common navigation sequences.
- Example: *“70% of users visit (Home → Category → Checkout).”*

3. Clustering

- Groups users with similar browsing behavior.
- Example: One cluster = “Casual Browsers”, another = “Frequent Buyers”.

4. Classification

- Assigns user sessions into predefined classes.
- Example: Predicting whether a user is a Potential Buyer or Just a Visitor.

5. Path Analysis (Graph Mining)

- Analyzes user navigation paths through a website as a graph.

- Example: Detecting most common paths leading to checkout page.

✓ Summary (Exam-Ready)

- Text Mining: Extracts patterns from unstructured text using NLP + ML.
- Applications: Sentiment analysis, spam filtering, document classification.
- Web Usage Analysis Techniques: Association rules, sequential patterns, clustering, classification, path analysis.

3.

(a) Define a data warehouse. List and explain all the 3 types of data warehouses.

Ans :

Definition of Data Warehouse

- A Data Warehouse (DW) is a centralized repository that stores integrated, subject-oriented, time-variant, and non-volatile data collected from multiple heterogeneous sources.
- It is designed to support querying, reporting, and decision-making rather than transaction processing.

👉 In simple words: A Data Warehouse is a decision support system (DSS) that provides a consolidated view of data for business intelligence and analytics.

Types of Data Warehouses

There are three main types of data warehouses depending on scope and usage:

1. Enterprise Data Warehouse (EDW)

- Definition:
A centralized warehouse that stores data for the entire organization.
- Characteristics:
 - Provides a unified view of data across all business areas.
 - Supports strategic decisions at the enterprise level.
 - Data is integrated and consistent.

- **Example:**

A multinational company maintaining one EDW that contains sales, finance, HR, and marketing data for all regions.

2. Data Mart

- **Definition:**

A subset of a data warehouse, usually oriented to a specific business line, department, or subject area.

- **Types:**

- **Dependent Data Mart:** Created from an existing EDW.
- **Independent Data Mart:** Built directly from operational systems.

- **Characteristics:**

- Smaller in scope and size.
- Faster to implement, less costly than full DW.

- **Example:**

A Sales Data Mart containing only sales-related information for quick analysis by the sales department.

3. Virtual Data Warehouse

- **Definition:**

A logical (view-based) data warehouse that provides a consolidated view of data without physically storing it in one repository.

- **Characteristics:**

- Provides real-time access to integrated data.
- Faster and cheaper to implement.
- May depend on middleware or query tools.

- **Example:**

A company using a virtual layer to integrate data from Oracle (finance), MySQL (sales), and MongoDB (customer data) without creating a central copy.

Summary (Exam-Ready)

- **Data Warehouse:** Centralized, subject-oriented, time-variant, non-volatile repository for decision support.
- **Types:**

1. Enterprise DW (EDW): Organization-wide, integrated.
2. Data Mart: Subset for a department/subject area (e.g., Sales, HR).
3. Virtual DW: Logical, real-time view without physical storage.

(b) What is data transformation ? List and explain all the data transformation steps.

Ans :

Definition of Data Transformation

- Data Transformation is a data preprocessing technique in which raw data is converted into a suitable format for storage in a data warehouse and for analysis in data mining.
- It ensures consistency, accuracy, and usability of data.
- Example: Converting *date* “12/08/25” into “2025-08-12 (YYYY-MM-DD)” format.

Steps in Data Transformation

The major steps involved are:

1. Smoothing

- Definition: Removes noise (errors, outliers) from data.
- Techniques: Binning, regression, clustering.
- Example: Replace noisy temperature readings (25, 26, 500, 27) with smoothed values (25, 26, 27).

2. Aggregation

- Definition: Summarizing or combining data.
- Use: Helps in creating data cubes for OLAP.
- Example: Daily sales data → aggregated into Monthly sales or Yearly sales.

3. Generalization

- Definition: Replacing low-level data with higher-level concepts using concept hierarchies.

- **Example:**
 - City → State → Country
 - Age 23 → Age Group “20–30”.

4. Normalization (Scaling)

- **Definition:** Adjusts numerical data into a common scale without distorting differences.
- **Techniques:**
 - Min-Max normalization
 - Z-score normalization
 - Decimal scaling
- **Example:** Income values (₱10,000 – ₱1,00,000) normalized to a range 0–1.

5. Attribute/Feature Construction

- **Definition:** Creating new attributes from existing ones to improve analysis.
- **Example:** From “Date_of_Birth”, construct a new attribute “Age”.

6. Discretization

- **Definition:** Converting continuous data into discrete categories.
- **Example:** Salary values → Low, Medium, High.

✓ Summary Table (Exam-Ready)

Step	Meaning	Example
Smoothing	Remove noise/outliers	500°C → replaced with avg 26°C
Aggregation	Summarize data	Daily Sales → Monthly Sales
Generalization	Replace with higher-level concepts	City → State → Country
Normalization	Scale values to a range	Income ₱10k–₱100k → 0–1

Attribute Construction	Create new attributes	DOB → Age
Discretization	Convert continuous → categorical	Salary → Low/Medium/High

✓ Final Answer (short form to write in exam):

- Data Transformation converts raw data into a usable format for data warehousing/mining.
- Steps: Smoothing, Aggregation, Generalization, Normalization, Attribute Construction, Discretization.

Qus 4.

(a) What is data mining ? List out the key differences between data warehousing and data mining. Mention any four applications of data mining.

Ans :

1. Definition of Data Mining

- Data Mining is the process of discovering patterns, correlations, trends, or useful knowledge from large datasets using techniques of statistics, machine learning, and database systems.
- It is also called Knowledge Discovery in Databases (KDD).
- Example: From supermarket transactions, discovering that *“70% of people who buy milk also buy bread.”*

2. Key Differences between Data Warehousing and Data Mining

Aspect	Data Warehousing	Data Mining
--------	------------------	-------------

Definition	Process of storing integrated, historical, and subject-oriented data from multiple sources in a centralized repository.	Process of analyzing large datasets to discover hidden patterns and relationships.
Purpose	Data storage and management for reporting and querying.	Knowledge discovery and prediction.
Focus	Organizing and consolidating data.	Extracting useful insights from data.
Techniques Used	ETL (Extract, Transform, Load), OLAP.	Machine Learning, Statistics, AI, Pattern Recognition.
Users	Managers, analysts (for decision support).	Data scientists, business analysts (for predictions).
Output	Clean, integrated data warehouse.	Actionable rules, clusters, predictions, patterns.

3. Applications of Data Mining (Any Four)

1. Market Basket Analysis

- Retailers find product associations (e.g., “*Diapers* → *Beer*” pattern).

2. Fraud Detection

- Banks and credit card companies detect unusual spending or fraudulent activities.

3. Customer Segmentation

- Telecom/marketing companies cluster customers for targeted campaigns.

4. Medical Diagnosis

- Helps doctors predict diseases by analyzing patient records.

5. (Optional Extra) Web Mining

- Analyzing web logs to understand user behavior (e.g., predicting e-commerce purchase patterns).

✓ Final Exam-Ready Answer

- **Data Mining:** Process of extracting hidden patterns, trends, and knowledge from large datasets.
- **Differences:** Warehousing = storage/integration, Mining = analysis/pattern discovery.
- **Applications:** Market Basket Analysis, Fraud Detection, Customer Segmentation, Medical Diagnosis.

(b) What is ETL ? Why do you need ETL in data warehouses ? Also, explain how ETL works in Data Warehousing ?

Ans :

1. Definition of ETL

- ETL stands for Extract, Transform, and Load.
- It is the core process of building a Data Warehouse, where data is collected from different sources, cleaned, transformed, and loaded into the warehouse for analysis.

👉 In simple words: ETL = “Taking data from source → Cleaning/Transforming → Storing into warehouse.”

2. Why Do We Need ETL in Data Warehouses?

We need ETL because:

1. Data comes from multiple sources (databases, ERP, CRM, flat files, web logs).
2. The data is often inconsistent, noisy, redundant.
3. ETL ensures:
 - Integration of heterogeneous data.
 - Cleansing & transformation into standard format.
 - Loading into the warehouse for OLAP & data mining.
4. Without ETL → warehouse would contain unclean, inconsistent, unusable data.

3. How ETL Works in Data Warehousing

Step 1: Extract

- Data is collected from different sources: relational DBs, legacy systems, spreadsheets, APIs, logs.
- Example: Extracting sales data from Oracle DB, customer info from MySQL, and transactions from CSV files.

Step 2: Transform

- Raw data is cleaned, standardized, and converted into a usable format.
- Operations include:
 - Data Cleaning (remove duplicates, handle missing values).
 - Data Transformation (normalization, generalization, aggregation).
 - Data Integration (resolve conflicts like *currency = USD vs INR*).
- Example: Convert dates to standard YYYY-MM-DD, normalize salaries, group customers by region.

Step 3: Load

- The transformed data is loaded into the Data Warehouse (fact tables, dimension tables).
- Loading can be:
 - Full Load (first time load).
 - Incremental Load (only new/updated records).
- Example: Load monthly sales into a Sales Fact Table linked with Product, Customer, and Time dimensions.

✓ Summary (Exam-Ready)

- ETL: Extract, Transform, Load – process to prepare and move data into Data Warehouse.
- Need: Ensures clean, consistent, integrated data for analysis.
- Working: Extract → Transform → Load → Stored in warehouse → Used for OLAP & Data Mining.

5. Write short notes on any four of the following :

(a) OLTP

(b) Data Marts**(c) Web structure mining****(d) Cloud Data Warehousing****(e) Data Granularity**

Ans :

(a) OLTP (Online Transaction Processing)

- **Definition:** OLTP refers to systems that manage day-to-day transactional data in real time.
- **Purpose:** Support routine business operations like insert, update, delete.
- **Characteristics:**
 - Large number of small transactions.
 - Normalized database design.
 - Response time must be very fast.
- **Example:** Banking systems, railway reservations, e-commerce order management.

(b) Data Marts

- **Definition:** A Data Mart is a subset of a Data Warehouse focused on a specific department or subject area.
- **Types:**
 1. Dependent Data Mart – Extracted from an enterprise data warehouse.
 2. Independent Data Mart – Built directly from operational data.
- **Advantages:** Faster access, less cost, easy to maintain.
- **Example:** Sales Data Mart, HR Data Mart, Marketing Data Mart.

(c) Web Structure Mining

- **Definition:** Web Structure Mining is the process of analyzing the link structure of the World Wide Web.
- **Goal:** Discover relationships among web pages and site structure.
- **Techniques Used:** Graph theory, PageRank, HITS algorithm.
- **Applications:**

- Search engine ranking (Google PageRank).
- Finding authoritative web pages.
- Website navigation optimization.

(d) Cloud Data Warehousing

- **Definition:** A Cloud Data Warehouse is a type of data warehouse hosted on cloud platforms.
- **Features:**
 - Scalable, flexible storage and computing.
 - Pay-as-you-go pricing.
 - Accessible from anywhere.
- **Examples:** Amazon Redshift, Google BigQuery, Snowflake.
- **Benefits:** Low cost, scalability, easy maintenance, integration with AI/ML tools.

(e) Data Granularity

- **Definition:** Data granularity refers to the level of detail or summarization in a data warehouse.
- **Types:**
 - Fine-grained data → More detail (e.g., sales per transaction).
 - Coarse-grained data → Summarized (e.g., monthly sales).
- **Importance:**
 - Fine granularity → greater detail, more storage required.
 - Coarse granularity → less storage, faster queries, but less detail.
- **Example:** Daily sales (fine-grained) vs yearly sales (coarse-grained).

✓ Final Tip for Exam

If the question says “*any four*”, choose OLTP, Data Marts, Web Structure Mining, and Data Granularity → these are easiest to explain with examples.

December, 2022

1.

(a) Define Dimensional Modelling. With reference to this, define the terms Facts, Fact Table, Dimensions and Dimensional Table. Give an example use-case and derive fact table and dimension tables.

Ans :

1. Dimensional Modelling – Definition

- **Dimensional Modelling (DM)** is a design technique in data warehousing used to structure data for query performance and easy analysis.
- **Data is organized into:**
 - **Fact Tables** (measurable numeric values)
 - **Dimension Tables** (contextual descriptive data)
- **DM is commonly used in Star Schema and Snowflake Schema designs.**

👉 **Goal:** To make data easy to understand for business users and analysts.

2. Key Terms in Dimensional Modelling

(a) Facts

- **Definition:** Numeric, measurable business data used for analysis.
- **Examples:** Sales amount, Quantity sold, Profit, Temperature, Score.
- **Property:** Facts are usually aggregated (SUM, AVG, COUNT).

(b) Fact Table

- **Definition:** A central table in dimensional modelling containing facts (measures) and foreign keys linking to dimension tables.
- **Characteristics:**
 - Contains quantitative values.
 - Usually very large.
- **Example Columns:** Date_ID, Product_ID, Store_ID, Sales_Amount, Quantity.

(c) Dimensions

- **Definition:** Descriptive attributes that give context to facts.

- **Examples:** Product name, Customer, Region, Time, Store.
- **Property:** Mostly textual, categorical data.

(d) Dimension Table

- **Definition:** A table that stores descriptive information about dimensions.
- **Characteristics:**
 - Usually smaller than fact tables.
 - Connected to fact table via primary key → foreign key relationship.
- **Example Columns:** Product_ID, Product_Name, Category, Brand.

3. Example Use Case – Retail Sales Analysis

A retail company wants to analyze sales performance.

Fact Table: Sales_Fact

Date_ID	Product_ID	Customer_ID	Store_ID	Sales_Amount	Quantity_Sold
D01	P01	C01	S01	5000	10
D02	P02	C02	S02	7000	14

Dimension Tables

(a) Date_Dimension

Date_ID	Date	Month	Quarter	Year
D01	2025-08-15	August	Q3	2025
D02	2025-08-16	August	Q3	2025

(b) Product_Dimension

Product_ID	Product_Name	Category	Brand
P01	Laptop	Electronics	Dell
P02	Smartphone	Electronics	Samsung

(c) Customer_Dimension

Customer_ID	Name	Region	Age_Group
C01	Ritesh	Kolkata	20–30
C02	Anjali	Delhi	30–40

(d) Store_Dimension

Store_ID	Store_Name	Location
S01	Store Kolkata	Kolkata
S02	Store Delhi	Delhi

✓ Summary (Exam-Ready)

- **Dimensional Modelling:** Organizes data into Fact Tables (measures) and Dimension Tables (descriptions).
- **Facts:** Quantitative measures (e.g., Sales Amount).

- **Fact Table:** Central table containing facts + foreign keys.
- **Dimensions:** Descriptive attributes (e.g., Product, Customer, Time).
- **Dimension Table:** Stores details of each dimension.
- **Example Use Case:** Retail sales → Fact Table (Sales_Fact) linked with Product, Customer, Store, and Time Dimensions.

(b) With the help of a Data Warehouse Architecture diagram, explain the following components and their significance :

(i) ETL

(ii) Metadata

(iii) Data Warehouse Access Tools

(iv) Data Warehouse Reporting Layer

Ans :

(i) ETL

(ii) Metadata

(iii) Data Warehouse Access Tools

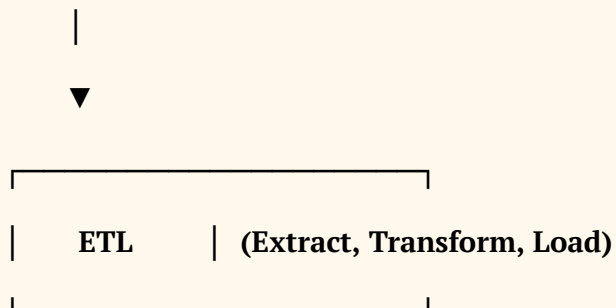
(iv) Data Warehouse Reporting Layer

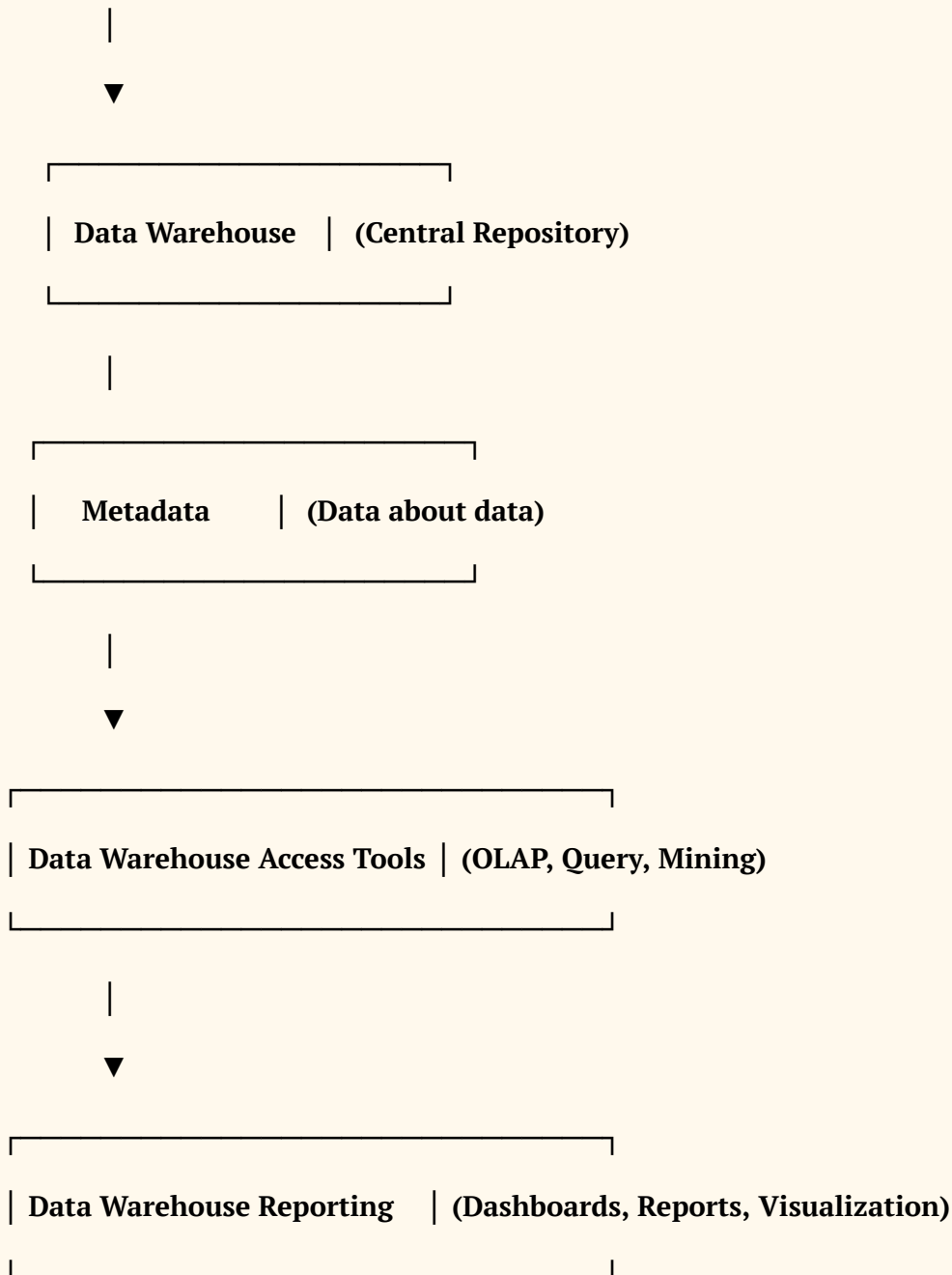
1. Data Warehouse Architecture Diagram

You can draw this simple layered diagram in exam 📌

[Operational Sources]

(DBs, ERP, CRM, Web logs, Flat files)





2. Components and Their Significance

(i) ETL (Extract, Transform, Load)

- **Definition:** The process of collecting data from multiple sources, cleaning & transforming it, and loading it into the data warehouse.

- **Functions:**
 - Extract → Collect data from databases, ERP, CRM, flat files, web logs.
 - Transform → Clean, normalize, aggregate, resolve inconsistencies.
 - Load → Store into fact & dimension tables of the data warehouse.
- **Significance:** Ensures that only clean, consistent, and integrated data is stored in the warehouse.

(ii) Metadata

- **Definition:** Metadata is data about data; it describes the structure, meaning, and rules of data in the warehouse.
- **Types:**
 - **Technical Metadata:** Schema definitions, data types, transformations, mappings.
 - **Business Metadata:** Business meaning of data, owner, usage rules.
- **Significance:** Acts as a roadmap for users and tools to understand and interpret warehouse data.

(iii) Data Warehouse Access Tools

- **Definition:** Software tools that allow end-users to query, analyze, and mine data in the warehouse.
- **Examples:**
 - OLAP tools → For multidimensional analysis (drill-down, roll-up, slice & dice).
 - Query tools → SQL-based querying.
 - Data Mining tools → Pattern discovery, clustering, classification.
- **Significance:** Enable decision-makers to interact with data for business insights.

(iv) Data Warehouse Reporting Layer

- **Definition:** The top layer that provides reports, dashboards, and visualizations to business users.
- **Examples:** Power BI, Tableau, Cognos, SAP BO.
- **Functions:**
 - Generate periodic reports (daily/weekly/monthly).
 - Provide ad-hoc analysis and KPIs.

- Present data visually via graphs, charts, dashboards.
- **Significance:** Helps business managers & executives take data-driven decisions.

✓ Final Exam-Ready Answer (Short Form)

- **ETL:** Extracts, cleans, and loads data into warehouse.
- **Metadata:** Data about data; roadmap for interpreting warehouse.
- **Access Tools:** OLAP, Query, Mining tools for analysis.
- **Reporting Layer:** Provides dashboards, reports, KPIs for decision-making.

(c) Define a Decision Tree. With the help of an example, explain the construction and representation of decision tree. Also, mention its strengths and weaknesses.

Ans :

1. Definition of Decision Tree

- A Decision Tree is a supervised machine learning model used for classification and prediction.
- It is a tree-structured model where:
 - **Internal nodes** → represent tests/attributes.
 - **Branches** → represent decision outcomes of the test.
 - **Leaf nodes** → represent final class labels or decisions.

👉 Used in classification because it converts complex decision-making into a set of simple rules.

2. Example – Constructing a Decision Tree

Problem: Predict whether a person will Play Tennis based on Weather conditions.

Training Data (simplified):

Outlook	Temperature	Wind	Play Tennis
---------	-------------	------	-------------

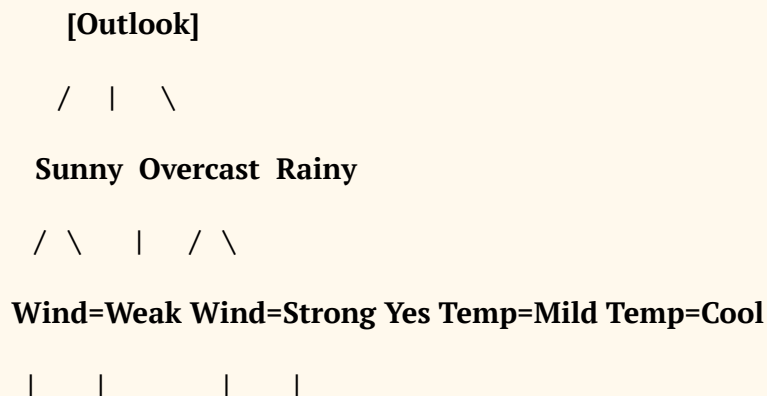
Sunny	Hot	Weak	No
-------	-----	------	----

Sunny	Mild	Strong	No
Overcast	Cool	Weak	Yes
Rainy	Mild	Weak	Yes
Rainy	Cool	Strong	No
Overcast	Hot	Strong	Yes

Decision Tree Construction (steps):

1. Choose root attribute using a splitting criterion (e.g., Information Gain / Gini Index).
 - Here, "Outlook" gives best split.
2. Split data based on values of attribute.
 - Outlook = Sunny → check Wind
 - Outlook = Rainy → check Temperature
 - Outlook = Overcast → Play = Yes
3. Continue recursively until all samples are classified or stopping criteria is met.

Representation of Decision Tree (Diagram)



Yes No Yes No

- **Interpretation:**
 - If Outlook = Overcast → Always Play.
 - If Outlook = Sunny and Wind = Weak → Play.
 - If Outlook = Sunny and Wind = Strong → Don't Play.
 - If Outlook = Rainy and Temp = Mild → Play, else No.

3. Strengths of Decision Trees

1. Easy to understand & interpret (flowchart-like).
2. Handles both categorical & numerical data.
3. No need for complex data preprocessing (scaling/normalization).
4. Can capture non-linear relationships.
5. Fast for small to medium datasets.

4. Weaknesses of Decision Trees

1. Overfitting – Trees can become too complex.
2. Sensitive to small changes – Minor change in data may lead to a different tree.
3. Biased towards attributes with more levels.
4. Not suitable for continuous prediction (better for categorical classification).

✓ Final Exam-Ready Answer (Summary)

- **Decision Tree:** A classification model with nodes (tests), branches (outcomes), and leaves (class labels).
- **Example:** "Play Tennis" prediction using attributes Outlook, Temperature, Wind.
- **Representation:** Tree diagram (root → branches → leaves).
- **Strengths:** Simple, interpretable, works on mixed data.
- **Weaknesses:** Overfitting, instability, biased towards multi-valued attributes.

(d) Discuss the following categories of Data Mining Issues :

(i) Mining Methodology and User Iteration Issues

(ii) Performance-based Issues

(iii) Diverse Data Types Issues

Ans :

(i) Mining Methodology and User Iteration Issues

These issues deal with how mining is carried out and how users interact with the system.

- **Choice of Data Mining Techniques:** Which algorithm (classification, clustering, association) is suitable?
- **Incorporating Domain Knowledge:** The system must allow inclusion of user knowledge to improve results.
- **Interactive Mining:** Users often need iterative refinement (query → result → feedback → re-run).
- **Interestingness of Results:** Ensuring that mined patterns are useful, novel, and understandable.
- **Handling Noisy / Incomplete Data:** Proper preprocessing methods (data cleaning, transformation) required.
- **Visualization Support:** Presenting results in user-friendly charts, graphs, dashboards.

👉 *Example:* In a medical dataset, doctors may iteratively refine classification rules to better diagnose diseases.

(ii) Performance-based Issues

These issues concern the efficiency, scalability, and speed of mining operations.

- **Scalability:** Mining algorithms should handle huge datasets (terabytes to petabytes).
- **Efficiency:** Optimized algorithms needed to process queries quickly.
- **Real-time Processing:** Some applications (fraud detection, stock market) require near real-time mining.
- **Parallel & Distributed Mining:** Use of cloud computing, parallel processors, Hadoop, Spark to handle big data.
- **Memory Management:** Efficient use of main memory and disk I/O to improve performance.

👉 **Example:** Fraud detection in credit card transactions requires fast, scalable mining to detect suspicious activities in real time.

(iii) Diverse Data Types Issues

Modern data is heterogeneous (not just structured tables). Mining must handle different formats.

- **Structured Data:** Traditional relational data (rows & columns).
- **Semi-Structured Data:** XML, JSON, logs.
- **Unstructured Data:** Text, images, audio, video, sensor data.
- **Spatial & Temporal Data:** GIS maps, satellite images, time-series stock data.
- **Web Data:** Web logs, clickstream, social media.
- **Integration Issues:** Combining data from multiple sources (databases, data lakes, APIs).

👉 **Example:** In e-commerce, customer data includes structured sales data, unstructured reviews, and images – mining must support all.

✅ Final Exam-Ready Answer (Summary)

1. Mining Methodology & User Iteration Issues → choice of algorithms, domain knowledge, user feedback, visualization.
2. Performance-based Issues → scalability, efficiency, real-time processing, distributed mining.
3. Diverse Data Types Issues → handling structured, semi-structured, unstructured, spatial, temporal, and web data.

2.

(a) Define Data Cleaning. Explain the ways and means of handling the Missing Values and Noisy Data while data preprocessing.

Ans :

1. Definition of Data Cleaning

- Data Cleaning is a data preprocessing technique used to improve the quality of data before analysis.
- It involves detecting and correcting (or removing) errors, inconsistencies, duplicates, missing values, and noise in data.
- Goal → To make data accurate, complete, and reliable for mining.

2. Handling Missing Values

Missing values occur when no data is recorded for a particular attribute.

Methods to handle them:

1. Ignore the tuple:
 - If the record has many missing attributes, drop the row.
 - ✓ Useful if dataset is large.
 - ✗ Risk: may lose important data.
2. Fill manually:
 - Domain expert/user fills in missing values.
 - ✓ Accurate, but time-consuming.
3. Global constant substitution:
 - Replace missing values with a constant like "Unknown" or "0".
 - ✓ Simple, but may mislead analysis.
4. Attribute mean/median/mode substitution:
 - Replace missing values with mean (numeric) or mode (categorical) of the attribute.
 - ✓ Preserves dataset size.
5. Predictive methods:
 - Use regression, decision tree, or machine learning models to estimate missing values.
 - ✓ More accurate, but computationally expensive.

👉 *Example:* In student records, if "Age" is missing, it may be replaced with the average age of students.

3. Handling Noisy Data

Noisy data = data with errors, outliers, or random variations.

Methods to handle them:

1. **Binning:**
 - Sort values and divide into bins; smooth using bin mean/median/boundaries.
 - Example: [21, 22, 23, 100] → anomaly 100 replaced by bin mean (say 22)
2. **Clustering:**
 - Group similar data; outliers (noise) lie outside clusters.
 - Example: Detecting fraudulent transactions.
3. **Regression:**
 - Fit data to a regression line and smooth deviations.
 - Example: Predicting house price trends.
4. **Outlier Analysis:**
 - Detect and remove extreme values using statistical methods (e.g., Z-score, IQR).
5. **Smoothing with Moving Averages:**
 - Replace each value with the mean of its neighborhood values.
 - Example: Stock market data smoothing.

✓ Final Exam-Ready Answer (Summary)

- **Data Cleaning:** Improves quality by handling missing and noisy data.
- **Missing Values Handling:** Ignore tuple, fill manually, use constants, mean/mode, or predictive models.
- **Noisy Data Handling:** Binning, clustering, regression, outlier detection, moving averages.

(b) Write and explain the K-means algorithm for clustering. How does it work ?

Ans :

1. Introduction to K-Means

- Clustering is an unsupervised learning technique used to group similar data points.
- K-Means Algorithm is one of the most popular clustering methods.

- It partitions a dataset into K clusters, where each cluster is represented by its centroid (mean).
- Objective → Minimize the sum of squared distances (SSD) between data points and their respective cluster centroids.

2. K-Means Algorithm (Steps)

Input: Number of clusters K , dataset of n objects.

Output: K clusters with centroids.

Steps:

1. **Initialize:** Randomly select K objects as initial cluster centroids.
2. **Assignment Step:** Assign each data point to the nearest centroid (using Euclidean distance).
3. **Update Step:** Recalculate the centroid of each cluster as the mean of all points in that cluster.
4. **Repeat:** Steps 2 and 3 until:
 - Centroids do not change, or
 - Maximum iterations reached.

3. Example – K-Means in Action

Suppose we have 6 data points in 2D:

(1,1), (2,1), (4,3), (5,4), (3,5), (6,6)

Let's cluster into $K = 2$ groups.

- **Step 1:** Randomly pick centroids: say (1,1) and (5,4).
- **Step 2:** Assign points:
 - Cluster 1: (1,1), (2,1)
 - Cluster 2: (4,3), (5,4), (3,5), (6,6)
- **Step 3:** Update centroids:
 - Cluster 1 mean = (1.5,1)
 - Cluster 2 mean = (4.5,4.5)
- **Step 4:** Repeat assignment & update until centroids stabilize.

Final:

- Cluster 1: small values (1,1), (2,1)
- Cluster 2: larger values (4,3), (5,4), (3,5), (6,6)

4. Working (Illustration)

The algorithm works as:

- Starts with random centroids → iterative refinement.
- Each iteration reduces intra-cluster distance and increases inter-cluster distance.
- Stops when clusters no longer change.

5. Strengths & Weaknesses of K-Means

Strengths:

- Simple and easy to implement.
- Efficient for large datasets.
- Produces tighter, spherical clusters.

Weaknesses:

- Requires pre-defining K (number of clusters).
- Sensitive to initial choice of centroids.
- Struggles with non-spherical clusters or clusters of different sizes.
- Sensitive to outliers and noise.

✓ Final Exam-Ready Answer (Summary)

- **K-Means:** An iterative clustering algorithm that partitions data into K clusters based on centroids.
- **Steps:** Initialize → Assign → Update → Repeat until stable.
- **Example:** 6 points clustered into 2 groups.
- **Strengths:** Simple, fast, scalable.
- **Weaknesses:** Needs K in advance, sensitive to initialization, poor for irregular clusters.

Qus 3.

(a) Why does dimensionality reduction of text need to be done ? Explain Tokenization process and Vector from text approach, with the help of a suitable example for each.

Ans :

1. Why Dimensionality Reduction of Text is Needed?

- In text mining, each unique word in a document can be considered a feature (dimension).
- For large corpora, the feature space may contain tens of thousands of words → high dimensionality problem.
- Issues with high dimensions:
 1. Sparsity → Most documents use only a small subset of vocabulary.
 2. Computational Complexity → More time and memory required.
 3. Noise & Irrelevance → Many words may not help classification (e.g., “the”, “is”, “a”).
 4. Curse of Dimensionality → Distance-based algorithms (like clustering) perform poorly.

👉 Dimensionality reduction helps by reducing vocabulary size → faster and more accurate models.

2. Tokenization Process

- Definition: Tokenization is the process of splitting text into smaller units (tokens) such as words, phrases, or sentences.
- It is the first step in text preprocessing.

Example:

Input sentence:

"Data Mining is an interesting field!"

Tokenized output:

["Data", "Mining", "is", "an", "interesting", "field"]

👉 Here, punctuation is removed and the sentence is broken into tokens.

- Later steps (like removing stopwords, stemming, lemmatization) further refine the tokens.

3. Vector from Text Approach

- After tokenization, text needs to be numerically represented (since algorithms work on numbers, not words).
- Vectorization converts tokens into feature vectors.

Common Approaches:

1. Bag of Words (BoW):

- Each document represented as a vector of word counts/frequencies.
- Order of words is ignored.

2. Example:

Documents:

- D1 = "Data Mining is fun"
- D2 = "Mining and Data Science"

3. Vocabulary = [Data, Mining, is, fun, and, Science]

- D1 vector = [1,1,1,1,0,0]
- D2 vector = [1,1,0,0,1,1]

2. TF-IDF (Term Frequency – Inverse Document Frequency):

- Adjusts word counts by importance (common words like “is” get lower weight).
- Example: If “Data” appears in all documents → lower weight. If “Science” appears rarely → higher weight.

3. Word Embeddings (Word2Vec, GloVe, BERT):

- Represent words as dense vectors capturing semantic meaning.
- Example: “King - Man + Woman ≈ Queen”.

✅ Final Exam-Ready Answer (Summary)

- **Dimensionality Reduction:** Needed in text mining to handle large vocabulary, reduce noise, improve performance.
- **Tokenization:** Splits text into tokens (e.g., “Data Mining is fun” → [Data, Mining, is, fun]).
- **Vector from Text:** Converts tokens into numerical form:
 - BoW/TF-IDF → sparse matrix of word frequencies.
 - Embeddings → dense vectors capturing semantic meaning.

(b) Enumerate the best practices for Data Warehouse Architecture.

Ans :

Designing a Data Warehouse (DW) requires careful planning to ensure scalability, performance, and usability. The following best practices are recommended:

1. Clear Business Requirements

- Define business objectives before designing DW.
- Focus on what kind of decisions the DW will support (sales trends, customer behavior, etc.).

2. Use a Layered Architecture

- Source Layer → OLTP, files, APIs.
- Staging Layer → ETL processes for cleaning and transformation.
- Data Warehouse Layer → fact tables, dimension tables.
- Data Mart Layer → subject-specific data (sales, finance).
- Presentation Layer → reporting, OLAP, dashboards.

3. Strong ETL Processes

- Efficient Extract-Transform-Load pipelines are essential.
- Handle missing data, noisy data, and duplicates properly.
- Automate ETL with scheduling & monitoring.

4. Metadata Management

- **Maintain technical metadata (data sources, transformations) and business metadata (definitions, KPIs).**
- **Ensures transparency and easy governance.**

5. Data Quality and Governance

- **Apply strict data cleaning, validation, and consistency checks.**
- **Define data ownership and access control policies.**

6. Scalability and Performance Optimization

- **Use partitioning, indexing, and compression to optimize queries.**
- **Consider cloud-based or distributed DW for big data.**
- **Implement incremental loads instead of full reloads.**

7. Security and Compliance

- **Apply role-based access control (RBAC).**
- **Ensure compliance with data privacy laws (GDPR, HIPAA, etc.).**
- **Encrypt sensitive data.**

8. Use of Data Marts

- **Build subject-specific Data Marts (finance, sales, HR).**
- **Supports faster and more focused queries.**

9. Regular Maintenance and Monitoring

- **Continuously monitor ETL jobs, query performance, and storage.**
- **Maintain logs for error tracking and recovery.**

10. Flexibility for Future Expansion

- **Design DW to handle new data sources, business needs, and analytical tools.**
- **Adopt modular and scalable architecture (e.g., cloud DW).**

 **Summary (Exam-Ready)**

Best practices in Data Warehouse Architecture include:

1. Clear business requirements.
2. Layered architecture design.
3. Strong ETL processes.
4. Metadata management.
5. Data quality & governance.
6. Scalability & performance optimization.
7. Security & compliance.
8. Use of Data Marts.
9. Regular monitoring & maintenance.
10. Flexibility for future expansion.

(c) Describe the following types of data marts :

(i) Dependent data marts

(ii) Independent data marts

Ans :

1. Data Mart – Quick Recap

- A Data Mart is a subset of a Data Warehouse, designed for a specific business area (e.g., Sales, Marketing, Finance).
- It allows faster access to focused data for decision-making.

There are two main types: Dependent and Independent.

2. (i) Dependent Data Marts

- A dependent data mart is built directly from an enterprise-wide Data Warehouse.
- The DW acts as the central source of truth, and the mart extracts its data from it.

Features:

- Data flows: Operational Sources → Data Warehouse → Dependent Data Mart.
- Ensures consistency across all departments (since data comes from a common warehouse).

- Easier to enforce data governance and standards.

Example:

- An organization has an enterprise Data Warehouse containing all company data.
- From it, a Sales Data Mart is created for sales managers and a Finance Data Mart for accountants.

👉 Both get their data from the DW, not directly from operational systems.

3. (ii) Independent Data Marts

- An independent data mart is created directly from operational or external data sources, without using a Data Warehouse.
- It is designed for a specific department's needs.

Features:

- Data flows: Operational Sources → Independent Data Mart.
- Quick to build, but may lead to data silos (inconsistencies between marts).
- Suitable for smaller organizations or when DW is not available.

Example:

- A Marketing Data Mart is built directly from customer relationship management (CRM) databases.
- A Sales Data Mart is separately built from ERP systems.
- They do not depend on a central warehouse.

4. Comparison Table

Feature	Dependent Data Mart	Independent Data Mart
Source of Data	Enterprise Data Warehouse	Directly from operational sources

Consistency	High (single source of truth)	Low (data silos possible)
Scalability	Highly scalable	Limited
Implementation	Slower, requires DW first	Faster, can be built standalone
Best For	Large organizations with DW	Small organizations / quick needs

✓ Summary (Exam-Ready)

- **Dependent Data Mart: Built from Data Warehouse → ensures consistency, centralized control.**
- **Independent Data Mart: Built directly from operational sources → quick, but may cause silos.**

Qus 4.

(a) With the help of an example use-case, explain the Snowflake schema. List its advantages and disadvantages.

Ans :

1. Definition of Snowflake Schema

- **A Snowflake Schema is a type of multidimensional data model used in data warehousing.**
- **It is an extension of the Star Schema, where dimension tables are normalized into multiple related tables.**
- **The schema looks like a snowflake, since dimension tables branch out into sub-dimensions.**

2. Example Use-Case

Suppose we are building a Sales Data Warehouse for a retail company.

Fact Table: Sales Fact

- Attributes: **Sale_ID, Date_Key, Product_Key, Customer_Key, Store_Key, Sales_Amount, Quantity_Sold.**

Dimension Tables (Normalized):

1. Product Dimension

- Product → linked to Sub-category → linked to Category.
- Example:
 - Product = “iPhone 15”
 - Sub-category = “Smartphones”
 - Category = “Electronics”

2. Customer Dimension

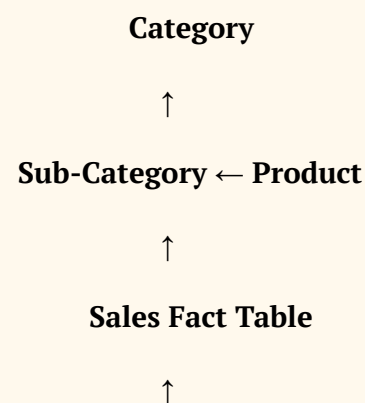
- Customer → linked to City → linked to State → linked to Country.
- Example:
 - Customer = “Ritesh”
 - City = “Howrah”
 - State = “West Bengal”
 - Country = “India”

3. Date Dimension

- Date → Month → Quarter → Year.

👉 This normalization reduces redundancy but creates multiple joined tables.

3. Snowflake Schema Diagram (Simple Representation)



Customer → City → State → Country

↑

Date → Month → Year

4. Advantages of Snowflake Schema

1. Normalization reduces redundancy → saves storage.
2. Better data integrity → since updates occur only in one place.
3. Supports complex queries across multiple levels of dimensions.
4. Scalable for large data models with multiple hierarchies.

5. Disadvantages of Snowflake Schema

1. Complexity → requires multiple joins, making queries slower compared to star schema.
2. Higher query execution time → more tables to join.
3. Harder to design and maintain compared to star schema.
4. Not as intuitive for end-users and BI tools.

✓ Summary (Exam-Ready)

- Snowflake Schema: A normalized form of star schema where dimension tables are split into multiple related tables.
- Example: Sales Fact with normalized Product, Customer, and Date dimensions.
- Advantages: Less redundancy, better integrity, scalability.
- Disadvantages: Complex, slower queries, harder to maintain.

(b) Define Web Mining. What kind of tasks can be performed using it ? Discuss its features and also a few applications.

Ans :

1. Definition of Web Mining

- Web Mining is the process of applying data mining techniques to discover useful patterns, trends, and knowledge from the World Wide Web.

- It extracts information from:
 - Web content (text, images, multimedia),
 - Web structure (links, page hierarchy), and
 - Web usage data (user logs, clickstream).

👉 Simply, Web Mining = Data Mining + Web Data

2. Tasks Performed using Web Mining

Some of the main tasks include:

1. **Web Content Mining** – Extracting knowledge from page contents (e.g., text mining on blogs, news sites).
2. **Web Structure Mining** – Analyzing hyperlink structure of websites (e.g., Google PageRank, HITS algorithm).
3. **Web Usage Mining** – Studying user behavior from server logs, cookies, clickstreams (e.g., recommendation systems).
4. **Personalization** – Providing tailored content to users (e.g., Amazon's product recommendations).
5. **Trend & Sentiment Analysis** – Understanding opinions and market trends from social media.

3. Features of Web Mining

- **Scalability:** Can process massive amounts of web data.
- **Heterogeneity Handling:** Works with structured (tables), semi-structured (HTML/XML), and unstructured (text, images) data.
- **Dynamic Data Support:** Can handle constantly updating web pages.
- **User-Centric:** Focuses on user behavior and preferences.
- **Integration:** Combines text mining, link analysis, and usage analysis.

4. Applications of Web Mining

1. **Search Engines:** Ranking and indexing (Google uses PageRank & HITS).
2. **E-commerce:** Recommendation systems (Amazon, Flipkart, Netflix).
3. **Marketing:** Customer segmentation and targeted advertising.
4. **Social Media Analysis:** Sentiment analysis, trending topics detection.
5. **Fraud Detection:** Identifying abnormal user behavior in online transactions.

6. E-learning: Adaptive content recommendation for learners.

✓ Summary (Exam-Ready)

- Web Mining = Data Mining applied on web data.
- Tasks: Web content, structure, and usage mining; personalization; trend analysis.
- Features: Scalable, handles heterogeneous & dynamic data, user-centric.
- Applications: Search engines, e-commerce, marketing, social media, fraud detection.

5. Write short notes on the following :

(a) Data Transformation (with reference to data preprocessing)

(b) Cloud Data Warehousing

(c) Data Lake and its Architecture

(d) Data Warehouse Automation

Ans :

(a) Data Transformation (with reference to Data Preprocessing)

- Definition: Data Transformation is a data preprocessing step where raw data is converted into a suitable format for analysis and mining.
- Purpose: Improve data quality, consistency, and compatibility.
- Steps involved:
 1. Smoothing – Removing noise from data (e.g., using binning).
 2. Aggregation – Summarizing data (e.g., daily → monthly sales).
 3. Normalization – Scaling values to a common range (e.g., min-max scaling).
 4. Generalization – Replacing low-level data with higher-level concepts (e.g., City → State → Country).
 5. Attribute/Feature Construction – Creating new derived attributes (e.g., BMI = weight/height²).

✓ Example: Transforming raw sales logs into structured monthly revenue data.

(b) Cloud Data Warehousing

- **Definition:** Cloud Data Warehousing is the practice of hosting a data warehouse on cloud platforms instead of on-premises infrastructure.
- **Examples:** Amazon Redshift, Google BigQuery, Snowflake.
- **Features:**
 - **Scalability:** On-demand storage and compute scaling.
 - **Cost-effective:** Pay-as-you-go model.
 - **Flexibility:** Supports structured & semi-structured data.
 - **Accessibility:** Accessible from anywhere with internet.
- **Advantages:** Reduced infrastructure cost, faster deployment, easy integration with cloud analytics tools.
- **Use-case Example:** A retail company uses Snowflake DW on AWS to store and analyze sales data across multiple regions.

(c) Data Lake and its Architecture

- **Definition:** A Data Lake is a central repository that stores raw, unstructured, semi-structured, and structured data at any scale.
- Unlike a Data Warehouse (structured, processed), a Data Lake stores data in its native/raw format.

Architecture Layers:

1. **Ingestion Layer** – Collects data from sources (databases, IoT, logs, APIs).
2. **Storage Layer** – Stores raw data in distributed systems (e.g., Hadoop, AWS S3).
3. **Processing Layer** – Cleans, transforms, and organizes data (using Spark, Hadoop, ETL tools).
4. **Consumption Layer** – Provides data access to BI tools, ML models, dashboards.
5. **Governance Layer** – Manages metadata, access, and security.

✓ **Example:** Banking sector using a Data Lake to store transaction logs, social media feeds, and customer feedback for fraud detection + analytics.

(d) Data Warehouse Automation (DWA)

- **Definition:** Data Warehouse Automation refers to the use of automation tools and techniques to design, build, deploy, and manage data warehouses faster.
- **Purpose:** Reduce manual coding, improve productivity, ensure consistency.

- **Features:**
 1. Automated schema generation (fact/dimension tables).
 2. Automated ETL pipeline creation.
 3. Built-in metadata management.
 4. Continuous integration & deployment (CI/CD) support.
- **Advantages:** Faster implementation, reduced human error, cost savings.
- **Example Tools:** WhereScape, TimeXtender, Talend, Azure Synapse DWA

✓ **Example:** A telecom company uses DWA to quickly integrate call records, billing data, and customer profiles into a warehouse.

✓ **Final Summary (Exam-Ready):**

- **Data Transformation:** Preprocessing step → smoothing, normalization, aggregation, generalization.
- **Cloud Data Warehousing:** DW hosted in cloud (Redshift, BigQuery, Snowflake) → scalable & cost-effective.
- **Data Lake:** Stores raw structured + unstructured data → architecture: ingestion, storage, processing, consumption, governance.
- **Data Warehouse Automation:** Use of automation tools for faster DW development, ETL, and deployment.

June 2023

1.

(a) Discuss the following data preprocessing stages briefly :

(i) Data Cleaning

(ii) Data Integration

(iii) Data Reduction

(iv) Data Transformation

Ans :

(i) Data Cleaning

- **Definition:** The process of detecting, correcting, or removing errors, inconsistencies, and missing values from data to improve quality.
- **Techniques:**
 - Handling missing values: Fill with mean/median, interpolation, or removal.
 - Noise removal: Smoothing, clustering, regression methods.
 - Consistency checking: Standardizing formats (e.g., "IN" vs "India").
- **Example:** If a customer database has *missing age values*, replace them with the average age.

(ii) Data Integration

- **Definition:** Combining data from multiple heterogeneous sources into a single unified view.
- **Issues handled:**
 - Schema integration: Aligning attribute names (e.g., "Cust_ID" vs "CustomerNo").
 - Data conflicts: Resolving differences in formats (e.g., "DD/MM/YYYY" vs "MM/DD/YYYY").
 - Entity identification: Ensuring one entity isn't duplicated across sources.

- **Example:** Integrating sales data from an ERP system with customer data from a CRM system.

(iii) Data Reduction

- **Definition:** Process of reducing the volume of data while maintaining its integrity and analytical value.
- **Techniques:**
 - **Dimensionality reduction:** PCA (Principal Component Analysis), attribute selection.
 - **Data compression:** Encoding, wavelet transforms.
 - **Aggregation:** Summarizing (e.g., daily → monthly sales).
 - **Sampling:** Choosing representative subsets.
- **Example:** Instead of storing transaction-level sales, store only monthly total sales.

(iv) Data Transformation

- **Definition:** Converting raw data into an appropriate format for mining/analysis.
- **Steps:**
 1. **Smoothing** – Removing noise.
 2. **Normalization** – Scaling values (e.g., 0–1 range).
 3. **Aggregation** – Summarizing values.
 4. **Generalization** – Replacing low-level data with higher concepts (City → State → Country).
 5. **Feature construction** – Creating new derived attributes.
- **Example:** Normalizing income values so that they lie in the range 0 to 1 before applying a clustering algorithm.

✓ Final Exam-Ready Summary

- **Data Cleaning:** Removes noise, missing values, inconsistencies.
- **Data Integration:** Combines heterogeneous data sources into a unified view.
- **Data Reduction:** Reduces data volume but keeps meaning (sampling, aggregation, PCA).

- **Data Transformation:** Converts data into suitable format (normalization, smoothing, generalization).

(b) Define Aggregate fact tables and derived dimension tables. What are their significance ? Give an example. Also mention their advantages and disadvantages.

Ans :

1. Aggregate Fact Tables

- **Definition:**
An aggregate fact table is a fact table that stores pre-computed summarized (aggregated) data instead of storing only detailed-level facts.
- **Purpose:** To speed up query performance by avoiding repeated aggregation during analysis.

 **Example:**

Detailed Fact Table → Stores daily sales data:

Date, Product, Store, Sales_Amount

01-01-2025, iPhone 15, Kolkata, 100000

01-01-2025, Samsung S23, Kolkata, 80000

Aggregate Fact Table → Stores monthly sales summary:

Month, Product_Category, Region, Total_Sales

Jan-2025, Smartphones, East, 18,00,000

2. Derived Dimension Tables

- **Definition:**
A derived dimension table is created by transforming, summarizing, or deriving attributes from an existing dimension table, instead of being directly loaded from the source system.
- **Purpose:** To provide higher-level abstraction or grouped categories.

✓ Example:

Original Customer Dimension:

Cust_ID, Name, Age, City, State, Country

Derived Dimension:

Age_Group (18-25, 26-40, 40+), Region (East, West, North, South)

3. Significance

- **Aggregate Fact Tables:**
 - Reduce query response time for analytical queries.
 - Enable faster OLAP operations (roll-up, drill-down).
- **Derived Dimension Tables:**
 - Simplify queries by providing predefined categories.
 - Improve usability by supporting hierarchies and roll-ups (e.g., City → State → Country).

4. Advantages and Disadvantages

Aggregate Fact Tables

✓ Advantages:

- Faster query performance (less on-the-fly aggregation).
- Useful for dashboard reporting and BI tools.
- Reduces CPU load on the warehouse.

✗ Disadvantages:

- Extra storage requirement.
- Data redundancy since details are also stored in the base fact table.
- Needs periodic refresh/update to remain consistent.

Derived Dimension Tables

✓ Advantages:

- Provide simplified, user-friendly categories.

- Reduce complexity in queries.
- Support hierarchies for drill-down analysis.

✗ Disadvantages:

- Extra processing to generate and maintain derived dimensions.
- May cause loss of granularity (e.g., grouping ages into ranges hides exact values).
- More ETL complexity.

✓ Final Exam-Ready Summary

- **Aggregate Fact Table:** Stores pre-computed summaries (e.g., monthly sales).
- **Derived Dimension Table:** Created by transforming existing dimensions (e.g., age → age groups).
- **Significance:** Improve query performance, simplify analysis, enable roll-ups.
- **Advantages:** Faster queries, user-friendly, supports OLAP.
- **Disadvantages:** Extra storage, redundancy, ETL complexity.

(c) Enumerate the key challenges in data warehouse design.

Ans :

Designing a Data Warehouse (DW) is a complex process involving integration of multiple data sources, ensuring scalability, and meeting business needs. The key challenges are:

1. Data Integration Challenge

- Combining data from heterogeneous sources (databases, flat files, ERP, CRM, web logs).
- **Issues:** Schema mismatches, different formats, duplicate records.
- **Example:** *Customer_ID stored as numeric in one system and alphanumeric in another.*

2. Data Quality Challenge

- Source data often contains errors, missing values, and inconsistencies.
- Poor data quality reduces trust in the warehouse.
- Requires data cleaning, deduplication, and standardization.

3. Scalability Challenge

- Data warehouses store large and continuously growing data volumes.
- Must scale for storage, performance, and concurrent user queries.

4. Performance Challenge

- OLAP queries are complex and compute-intensive (aggregations, drill-downs).
- Need optimization using indexes, partitioning, aggregate fact tables, caching

5. ETL (Extract-Transform-Load) Complexity

- ETL process consumes time and resources.
- Must handle:
 - Incremental loads (new data).
 - Data transformation (e.g., currency conversion, date standardization).
 - Error handling.

6. Metadata Management Challenge

- Metadata (data about data) must track source, transformations, definitions.
- Without proper metadata, users cannot understand or trust reports.

7. Security and Privacy Challenge

- Warehouses contain sensitive business data.
- Must implement role-based access control, encryption, auditing.

8. High Cost and Time Challenge

- Designing, building, and maintaining a DW requires significant investment in hardware, software, and skilled personnel.
- ROI (Return on Investment) may take time

9. Evolving Business Requirements

- Business needs change frequently → DW schema may need modifications.

- Difficult to balance flexibility vs. stability of the design.

✓ Final Summary (Exam-Ready Points)

Key challenges in data warehouse design are:

1. Data Integration
2. Data Quality
3. Scalability
4. Performance
5. ETL Complexity
6. Metadata Management
7. Security & Privacy
8. High Cost & Time
9. Evolving Business Requirements

(d) Differentiate between a data lake and a data warehouse.

Ans :

1. Definition

- **Data Lake:**
A central repository that stores raw, unprocessed, structured, semi-structured, and unstructured data at scale.
- **Data Warehouse:**
A repository that stores processed, cleaned, structured data optimized for reporting and analysis.

2. Data Type

- **Data Lake:** Structured (tables), semi-structured (JSON, XML), unstructured (images, videos, logs).
- **Data Warehouse:** Mostly structured and some semi-structured data.

3. Storage Approach

- **Data Lake:** *Schema-on-read* → Data is stored first (raw) and schema is applied when used.
- **Data Warehouse:** *Schema-on-write* → Data is transformed and structured before storage.

4. Purpose

- **Data Lake:** For data exploration, machine learning, big data analytics.
- **Data Warehouse:** For reporting, OLAP, dashboards, business intelligence.

5. Users

- **Data Lake:** Data Scientists, Analysts, ML Engineers.
- **Data Warehouse:** Business Analysts, Managers, Decision Makers.

6. Processing Framework

- **Data Lake:** Uses big data frameworks like Hadoop, Spark.
- **Data Warehouse:** Uses RDBMS/OLAP engines (e.g., Oracle, Teradata, Redshift).

7. Cost & Storage

- **Data Lake:** Low-cost storage (HDFS, AWS S3).
- **Data Warehouse:** Higher cost due to preprocessing and indexing.

8. Example

- **Data Lake:** *A bank storing raw transaction logs, social media data, and call center recordings in Hadoop HDFS.*
- **Data Warehouse:** *The same bank storing structured customer transaction summaries in Oracle DW for reporting.*

✓ Comparison Table: Data Lake vs Data Warehouse

Aspect

Data Lake 

Data Warehouse 

Data Type	Raw, structured, semi-structured, unstructured	Structured, processed
Schema	Schema-on-read	Schema-on-write
Purpose	ML, data science, advanced analytics	Reporting, BI, OLAP
Users	Data scientists, engineers	Business analysts, managers
Storage	Low-cost (HDFS, S3)	Costly (RDBMS, OLAP DB)
Performance	Slower for queries (raw data)	Optimized for queries
Flexibility	Very flexible	Rigid, predefined schema
Examples	Hadoop, AWS S3, Azure Data Lake	Oracle, Teradata, Snowflake

✓ Final Summary (Exam-Ready)

- Data Lake = Raw + flexible + cheap storage + schema-on-read.
- Data Warehouse = Processed + structured + optimized for BI + schema-on-write.

(e) Write and explain the Apriori algorithm to identify the most frequently occurring elements and meaningful associations in a dataset.

Ans :

1. Introduction

- The Apriori Algorithm is a classic association rule mining algorithm used in Market Basket Analysis.
- It helps to find frequent itemsets (items frequently bought together) and then derive association rules (meaningful relationships among items).

👉 **Example:** In a supermarket, $\{Milk\} \rightarrow \{Bread\}$ means customers who buy milk also tend to buy bread.

2. Working Principle (Apriori Property)

- Apriori uses the principle:
“If an itemset is frequent, then all its subsets must also be frequent.”
- This property reduces the search space efficiently.

3. Steps of the Apriori Algorithm

1. **Input:** Transaction dataset, minimum support, minimum confidence.
2. **Generate Candidate Itemsets (C_k):** List of all possible item combinations of size k .
3. **Prune using Apriori Property:** Remove candidates whose subsets are not frequent.
4. **Count Support:** Check frequency of candidates in the dataset.
5. **Form Frequent Itemsets (L_k):** Keep itemsets with support $\geq$ minimum support.
6. **Repeat:** Increase k until no more frequent itemsets are found.
7. **Generate Association Rules:** From frequent itemsets, generate rules that satisfy minimum confidence

4. Pseudocode of Apriori Algorithm

Apriori(D, min_support)

1. $L_1 = \{\text{frequent items}\}$
2. **for** ($k = 2$; $L_{k-1} \neq \emptyset$; $k++$) **do**
3. $C_k = \text{generate_candidates}(L_{k-1})$
4. **for each** transaction t in D **do**
5. count support of candidates in C_k
6. **end for**
7. $L_k = \text{candidates in } C_k \text{ with support } \geq \text{min_support}$
8. **end for**
9. **Return Union of all L_k**

5. Example

Transaction Dataset:

TID	Items Purchased
T1	Milk, Bread, Butter
T2	Milk, Bread
T3	Bread, Butter
T4	Milk, Bread, Butter, Jam
T5	Milk, Butter

Step 1: Minimum Support = 60% (3 out of 5 transactions).

- L1 (Frequent 1-itemsets): {Milk (4), Bread (4), Butter (4)}
- Jam occurs only once → removed.

Step 2: Generate L2 (Frequent 2-itemsets):

- {Milk, Bread} (3), {Milk, Butter} (3), {Bread, Butter} (3) → Frequent.

Step 3: Generate L3 (Frequent 3-itemsets):

- {Milk, Bread, Butter} (2) → Not frequent (only 40%).

✓ Final Frequent Itemsets: {Milk}, {Bread}, {Butter}, {Milk, Bread}, {Milk, Butter}, {Bread, Butter}.

Step 4: Generate Association Rules:

- {Milk} → {Bread}, {Bread} → {Milk}, {Milk, Butter} → {Bread}, etc.

6. Applications

- Market basket analysis (supermarkets, e-commerce).
- Web usage mining (pages visited together).
- Healthcare (symptoms → disease).
- Telecom (service bundles).

7. Advantages & Limitations

✓ Advantages:

- Simple and easy to implement.
- Reduces search space using Apriori property.

✗ Limitations:

- Generates a large number of candidate sets.
- Computationally expensive for very large datasets.

✓ Final Exam-Ready Answer

- Apriori Algorithm: Finds frequent itemsets and generates association rules.
- Steps: Candidate generation → Pruning → Support counting → Frequent itemsets → Rule generation.
- Example: From supermarket transactions, Apriori finds rules like $\{Milk\} \rightarrow \{Bread\}$.
- Use: Widely applied in market basket analysis, web mining, healthcare, and telecom.

Qus 2.

(a) What is Cluster Analysis ? How is this used in Data Mining ? Give an example. Also mention few applications of cluster analysis in data mining.

Ans :

1. Definition

- **Cluster Analysis (or Clustering)** is a data mining technique that groups a set of objects into clusters (groups) such that:
 - Objects in the same cluster are more similar to each other.
 - Objects in different clusters are more dissimilar.

👉 It is an unsupervised learning method (no predefined labels).

2. Role in Data Mining

- Cluster analysis is used in data mining to:
 - Identify natural groupings within large datasets.
 - Discover hidden patterns without prior knowledge.
 - Reduce data complexity by grouping similar records.
 - Preprocess data for classification, prediction, and anomaly detection.

3. Example

Suppose we have customer data with attributes: Age, Income, Spending Score.

Using clustering:

- Cluster 1 → Young, high-spending customers.
- Cluster 2 → Middle-aged, moderate-spending customers.
- Cluster 3 → Older, low-spending customers.

👉 A retail store can use these clusters for targeted marketing.

4. Applications of Cluster Analysis in Data Mining

1. **Market Segmentation**
 - Grouping customers based on buying behavior.
 - Example: Amazon groups customers for personalized recommendations.
2. **Image Segmentation**
 - Dividing an image into regions (clusters of pixels).
 - Example: Medical imaging to detect tumors.
3. **Document/Text Clustering**
 - Grouping news articles, blogs, or research papers into topics.
4. **Anomaly Detection**

- Identifying outliers such as fraudulent credit card transactions.
- 5. Bioinformatics
 - Grouping genes with similar expression patterns.

✓ Final Exam-Ready Answer

- Cluster Analysis = Grouping similar objects into clusters.
- Use in Data Mining = Helps in pattern discovery, market segmentation, anomaly detection.
- Example = Customers grouped based on age, income, and spending for targeted marketing.
- Applications = Market segmentation, image processing, document clustering, fraud detection, bioinformatics.

(b) List and discuss various types of Webmining.

Ans :

1. Definition

- Web Mining is the process of discovering useful patterns, knowledge, and insights from web data (content, structure, and usage).
- Web data sources:
 - Web pages (text, multimedia).
 - Hyperlinks between pages.
 - Web server logs (user navigation).

2. Types of Web Mining

Web mining is broadly classified into three main types:

(i) Web Content Mining

- Extracting useful information from web page content (text, images, videos, audio, metadata).
- Techniques: Information Retrieval, Natural Language Processing (NLP), Text Mining.
- Example:

- Google indexing web pages based on keywords.
- Sentiment analysis of product reviews on Amazon.

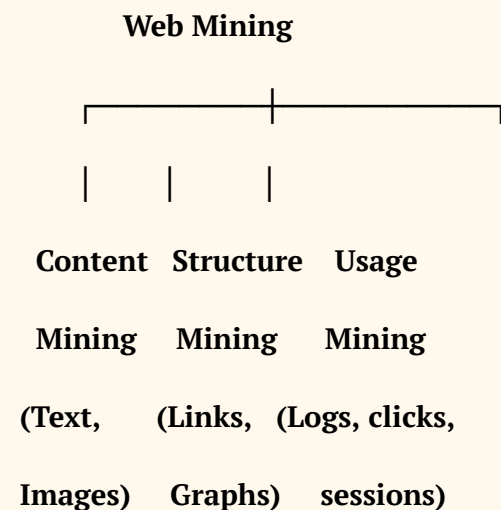
(ii) Web Structure Mining

- Analyzing the structure of hyperlinks between web pages (graph theory-based).
- Helps in identifying important/authoritative pages.
- Algorithms used: PageRank, HITS.
- Example:
 - Google uses PageRank to rank search results.
 - Identifying communities of related websites.

(iii) Web Usage Mining

- Analyzing user behavior through web logs, cookies, and clickstreams.
- Focuses on navigation patterns, session details, and user preferences.
- Example:
 - Amazon recommending products based on browsing history.
 - E-commerce sites analyzing abandoned shopping carts.

3. Diagram (Simple Representation)



4. Applications of Web Mining

- Personalized recommendations (Amazon, Netflix).
- Search engines (Google ranking using PageRank).

- Fraud detection (unusual browsing/purchase patterns).
- Customer profiling (market segmentation from browsing logs).

✓ Final Exam-Ready Summary

- Web Mining = Extracting knowledge from web data.
- Types:
 1. Web Content Mining → Extracting useful info from page content.
 2. Web Structure Mining → Analyzing hyperlinks between pages.
 3. Web Usage Mining → Studying user behavior via logs/clickstreams.

Qus 3.

(a) Explain the following techniques for Dimensionality Reduction :

(i) Feature Selection

(ii) Feature Extraction

Ans :

1. Introduction

- Dimensionality Reduction = Process of reducing the number of input variables (features) in a dataset while retaining as much useful information as possible.
- Needed because:
 - High-dimensional data increases storage, computation cost, and risk of overfitting.
 - Helps in data visualization and faster model training.

(i) Feature Selection

- Definition: Selecting a subset of the most relevant features from the original dataset, while removing redundant or irrelevant ones.
- Goal: Keep only features that contribute significantly to prediction/analysis.
- Techniques used:
 - Filter Methods (e.g., Chi-square test, Information Gain, Correlation).
 - Wrapper Methods (e.g., Forward Selection, Backward Elimination).

- Embedded Methods (e.g., Decision Trees, LASSO).

✓ Example:

Suppose you have features [Age, Height, Weight, EyeColor, Income, BloodPressure] to predict “Heart Disease”.

- “EyeColor” has no relation → removed.
- Final selected features: [Age, Income, BloodPressure].

(ii) Feature Extraction

- **Definition:** Transforming the original features into a new set of reduced features (lower dimensions) while retaining maximum variance/information.
- Instead of removing, it creates new features from old ones.
- **Techniques used:**
 - Principal Component Analysis (PCA) → creates new features (principal components).
 - Linear Discriminant Analysis (LDA).
 - Autoencoders (Neural Networks).

✓ Example:

In an image dataset (with thousands of pixels as features):

- PCA transforms pixel data into a few “principal components” that capture most of the image information.
- Instead of using all pixels, only 50–100 new extracted features are used.

3. Key Difference

Aspect	Feature Selection	Feature Extraction
Approach	Keeps subset of original features	Creates new transformed features
Data Meaning	Original meaning preserved	Original meaning lost (new features are combinations)

Techniques	Filter, Wrapper, Embedded	PCA, LDA, Autoencoders
Example	Remove "EyeColor"	PCA reduces 1000 features → 20 components

✓ Final Exam-Ready Answer

- Feature Selection = Choosing the most relevant subset of existing features (e.g., selecting Age & Income for prediction).
- Feature Extraction = Creating new features from existing ones (e.g., PCA converting pixel values into principal components).
- Both reduce dimensionality, improve efficiency, and help avoid overfitting.

(b) Discuss the layered implementation of ETL in a Data Warehouse.

Ans :

1. Introduction

- ETL (Extract, Transform, Load) is the process of extracting data from multiple sources, transforming it into a consistent format, and loading it into a Data Warehouse.
- ETL is usually implemented in layers, where each layer performs a specific function before passing data to the next layer.

2. Layered Implementation of ETL

The ETL process in a Data Warehouse can be organized into the following layers:

(i) Data Extraction Layer

- Responsible for collecting data from heterogeneous sources:
 - Relational databases (Oracle, MySQL).
 - Flat files (CSV, Excel).
 - Web APIs, ERP, CRM systems.
- Ensures data acquisition and transfer into the staging area.

- **Techniques:** Full extraction, Incremental extraction (delta loads).

✓ **Example:** Extracting sales data from ERP and customer data from CRM.

(ii) Staging Layer

- Acts as a temporary storage area where extracted data is kept before transformation.
- **Purpose:** Data validation, cleansing, and profiling.
- Helps detect and fix errors, duplicates, and missing values before loading into warehouse.

✓ **Example:** Removing duplicate customer records or invalid transaction IDs.

(iii) Transformation Layer

- Core layer where data is transformed into a standard format.
- **Tasks include:**
 - Data Cleaning (handle missing/noisy data).
 - Data Integration (merge from multiple sources).
 - Data Transformation (aggregation, standardization, normalization).
 - Business Rule Application (e.g., currency conversion, date formatting).

✓ **Example:** Converting all sales into INR, or mapping product codes to standard IDs.

(iv) Loading Layer

- Loads the transformed data into the Data Warehouse.
- **Can be:**
 - Full Load: Replace entire data.
 - Incremental Load: Load only new/changed data.
- Ensures data integrity, indexing, and partitioning for fast queries.

✓ **Example:** Daily incremental loading of new transactions into the fact table.

(v) Data Warehouse Layer

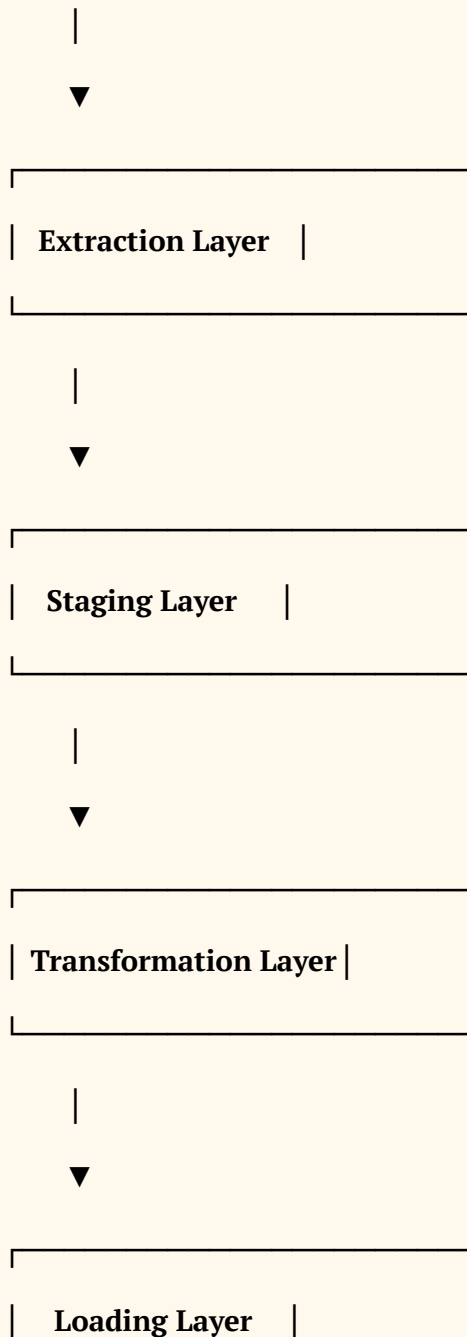
- Final structured storage of data in Fact Tables and Dimension Tables (using star or snowflake schema).

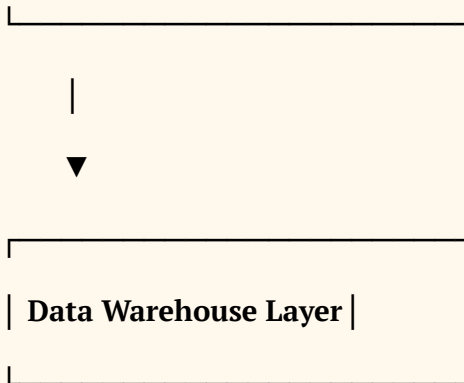
- Optimized for OLAP queries, reporting, and analysis.

3. ETL Layered Architecture Diagram (textual sketch for exam)

Source Systems

(Databases, Files, APIs)





4. Benefits of Layered ETL Implementation

- Improves data quality and consistency.
- Provides a systematic workflow for handling large data.
- Makes ETL modular and manageable.
- Easier to debug, monitor, and optimize performance.

✓ Final Exam-Ready Answer

- ETL in a Data Warehouse is implemented in layers:
 1. Extraction Layer – collects data from sources.
 2. Staging Layer – stores and validates extracted data.
 3. Transformation Layer – cleans, integrates, and standardizes data.
 4. Loading Layer – loads transformed data into the warehouse.
 5. Data Warehouse Layer – stores data for OLAP queries and reporting.
- This layered approach ensures data quality, consistency, and performance optimization.

Qus 4.

(a) Define OLAP. Differentiate between Multicube and Hypercube. Mention the applications of OLAP reporting system.

Ans :

1. Definition of OLAP

- OLAP (Online Analytical Processing) is a category of software tools that enables analysts, managers, and executives to gain insights by interactively analyzing multidimensional data.
- OLAP systems support complex queries, aggregations, and drill-down operations on large volumes of historical data in a Data Warehouse.
- They are optimized for decision-making, not transactional processing (unlike OLTP).

✓ **Example:** A retail company using OLAP to analyze sales by region, product, and time.

2. Differentiation: Multicube vs Hypercube

Aspect	Multicube	Hypercube
Definition	A collection of smaller cubes, each representing a specific business process or subject area.	A single multidimensional cube that represents all dimensions and facts together.
Data Structure	Multiple cubes linked together but stored separately.	One unified cube with multiple dimensions.
Flexibility	More flexible – different cubes can be designed for different business needs.	Less flexible – one cube must support all requirements.
Complexity	Easier to design and maintain since each cube is smaller.	More complex as dimensions grow (curse of dimensionality).
Example	A retail company having separate cubes: Sales Cube, Inventory Cube, Customer Cube.	A single cube having Product × Region × Time × Customer × Supplier dimensions.

3. Applications of OLAP Reporting System

OLAP reporting systems are widely used for business intelligence and decision support:

1. **Business Performance Analysis**
 - Sales forecasting, profitability analysis, performance dashboards.
 - *Example:* Analyzing sales by product, region, and month.
2. **Financial Reporting**
 - Budgeting, financial planning, variance analysis.
3. **Marketing & Customer Relationship Management (CRM)**
 - Customer segmentation, campaign effectiveness, churn analysis.
4. **Supply Chain & Inventory Management**
 - Demand forecasting, stock-level monitoring, vendor performance.
5. **Healthcare & Education**
 - Patient treatment pattern analysis, student performance monitoring.
6. **Government & Public Sector**
 - Fraud detection, tax analysis, census data evaluation.

✓ Final Exam-Ready Answer

- OLAP = Online Analytical Processing; used for multidimensional analysis of data stored in Data Warehouses.
- Multicube vs Hypercube:
 - *Multicube* = multiple smaller cubes for different business processes.
 - *Hypercube* = one single cube with multiple dimensions.
- Applications of OLAP Reporting: Business analysis, financial planning, marketing, supply chain optimization, healthcare monitoring, fraud detection.

(b) List and explain the following types of Data Warehouses :

(i) Enterprise Data Warehouse (EDW)

(ii) Operational Data Store

Ans :

(i) Enterprise Data Warehouse (EDW)

- **Definition:**
An Enterprise Data Warehouse is a centralized and integrated repository of data collected from across the entire organization.
- It stores historical and current data, and is optimized for OLAP queries, reporting, and strategic decision-making.
- **Key Features:**
 1. **Subject-oriented** → Organized around key subjects like sales, customers, finance.
 2. **Integrated** → Combines data from multiple heterogeneous sources.
 3. **Time-variant** → Maintains historical data for trend analysis.
 4. **Non-volatile** → Data is stable (not frequently updated like OLTP systems).
- **Example:**
A large retail company building an EDW that integrates sales, inventory, HR, and finance data into one central repository for enterprise-wide analytics.

(ii) Operational Data Store (ODS)

- **Definition:**
An Operational Data Store is a real-time or near real-time repository that integrates data from multiple operational systems (OLTP systems).
- It is mainly used for operational reporting and short-term decision-making, not long-term historical analysis.
- **Key Features:**
 1. **Current/near real-time data** → Unlike EDW, ODS stores up-to-date information.
 2. **Integrated view** → Consolidates data from different operational systems.
 3. **Intermediate layer** → Often acts as a staging area before loading data into the EDW.
 4. **Supports operational decisions** → Helps managers and customer service teams.
- **Example:**
A bank's ODS that shows the latest customer account balances and transactions to support real-time customer queries.

✓ **Final Answer (Concise for Exams):**

- **Enterprise Data Warehouse (EDW):** A centralized, integrated data warehouse storing historical and current data across the entire organization; used for enterprise-wide analytics and strategic decisions.
- **Operational Data Store (ODS):** A real-time or near real-time integrated data store used for operational reporting, day-to-day decision-making, and as a staging area for EDW.

5. Write short notes on the following :

- (a) ELT vs. ETL**
- (b) Data Marts**
- (c) Applications of Data Mining**
- (d) OLAP data cube operations**

Ans :

(i) Enterprise Data Warehouse (EDW)

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December, 2023

1. (a) With the help of a diagram, describe the Conceptual Architecture of Hadoop Data Warehouse.

Ans :

1. Introduction

- A Hadoop Data Warehouse is a modern data warehouse that combines Hadoop's distributed storage (HDFS) and processing frameworks (MapReduce, Spark) with data warehouse features (ETL, metadata, reporting).
- It supports structured, semi-structured, and unstructured data at massive scale.

2. Conceptual Architecture of Hadoop Data Warehouse

The architecture is usually layered as follows:

(i) Data Sources Layer

- Multiple data sources feed into Hadoop:
 - Relational databases (OLTP systems).
 - Enterprise systems (ERP, CRM).
 - Log files, IoT devices, clickstream data.
 - Social media, sensors, external APIs.

(ii) Data Ingestion Layer

- Responsible for collecting and loading data into Hadoop Distributed File System (HDFS).
- Tools: Sqoop (for RDBMS to Hadoop), Flume/Kafka (for log/event streaming).

(iii) Hadoop Storage Layer (HDFS)

- HDFS (Hadoop Distributed File System) stores structured, semi-structured, and unstructured data.
- Data is replicated across nodes for fault tolerance and scalability.

(iv) Processing & ETL Layer

- **Data transformation and analytics using:**
 - **MapReduce, Apache Spark – distributed processing engines.**
 - **Hive, Pig – high-level query and scripting tools.**
- **Cleans, integrates, and transforms raw data into usable formats.**

(v) Metadata & Resource Management Layer

- **Hive Metastore → Stores schema, table definitions, partitions.**
- **YARN (Yet Another Resource Negotiator) → Manages cluster resources and job scheduling.**

(vi) Data Warehouse Layer

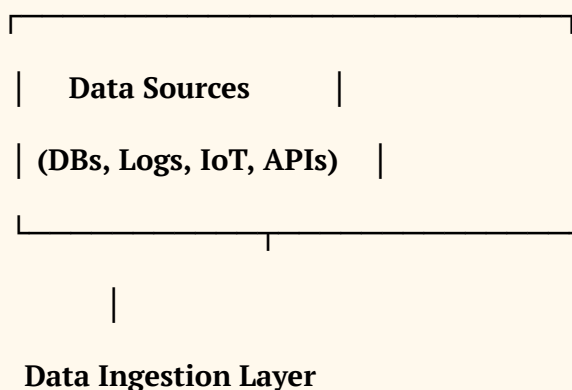
- **Provides schema-on-read (Hive, Impala) for querying big data as if it were in a warehouse.**
- **Stores processed data in fact and dimension tables for analysis.**

(vii) BI / Analytics / Access Layer

- **Business users and analysts interact with Hadoop warehouse via:**
 - **OLAP tools, BI dashboards (Tableau, Power BI, Pentaho).**
 - **SQL-on-Hadoop tools (HiveQL, Presto, Impala).**
- **Provides reporting, visualization, machine learning, and predictive analytics.**

3. Conceptual Architecture Diagram

Here's a textual diagram you can easily redraw in your exam:



(Squoop, Flume, Kafka)

|



| **Hadoop Storage (HDFS)** |

|

| **Processing & ETL Layer** |

| **(MapReduce, Spark, Hive, Pig)** |

|

| **Metadata & Resource Mgmt (Hive, YARN)** |

|

| **Data Warehouse Layer** |

| **(Hive Tables, Impala)** |

|

| **BI & Analytics Layer** |

| (Dashboards, OLAP, ML) |

4. Significance

- **Scalability:** Handles petabytes of data.
- **Flexibility:** Works with structured + unstructured data.
- **Cost-effective:** Open-source, commodity hardware.
- **Analytics-ready:** Integrates BI tools with Big Data.

✓ **Final Answer (Concise for Exams):**

The Hadoop Data Warehouse architecture consists of:

1. **Data Sources Layer** → input from DBs, logs, IoT, APIs.
2. **Ingestion Layer** → Sqoop, Flume, Kafka move data into Hadoop.
3. **Storage Layer (HDFS)** → distributed, fault-tolerant storage.
4. **Processing & ETL Layer** → MapReduce, Spark, Hive transform data.
5. **Metadata & Resource Management** → Hive Metastore, YARN.
6. **Data Warehouse Layer** → Hive/Impala provide schema and OLAP support.
7. **BI/Analytics Layer** → Reports, dashboards, ML, decision-making.

(b) Draw and explain star schema diagram and snow-flake schema diagram for the dimensions (Products, Customers, Time, Locations) and fact (Sales-Items) for the measures namely Quantity-sold and Amount-sold for a manufacturing company data warehouse dimensional modeling

Ans :

1. Problem Statement

We need to design a dimensional model for a manufacturing company with:

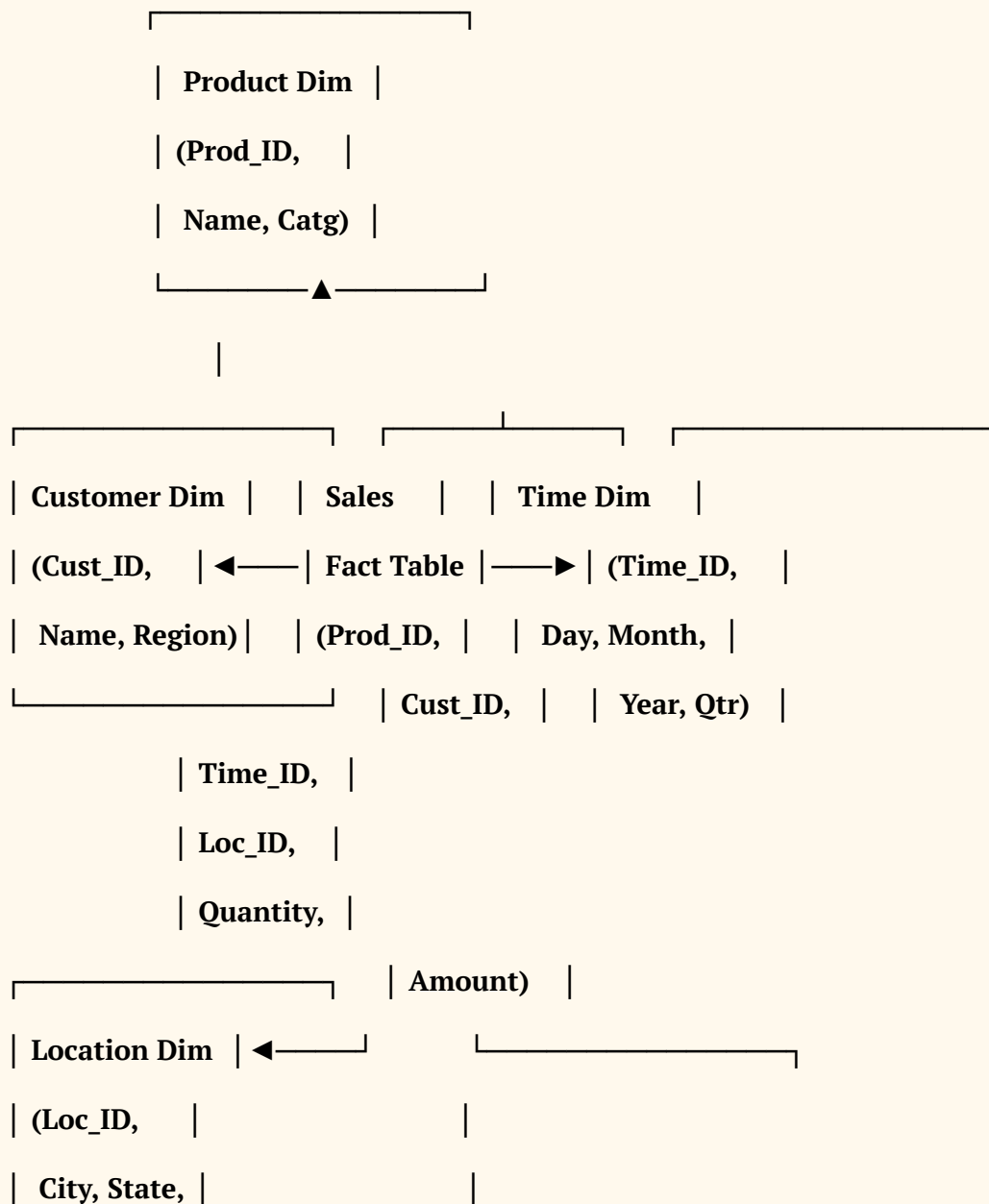
- **Dimensions** → Products, Customers, Time, Locations
- **Fact** → Sales-Items
- **Measures** → Quantity-sold, Amount-sold

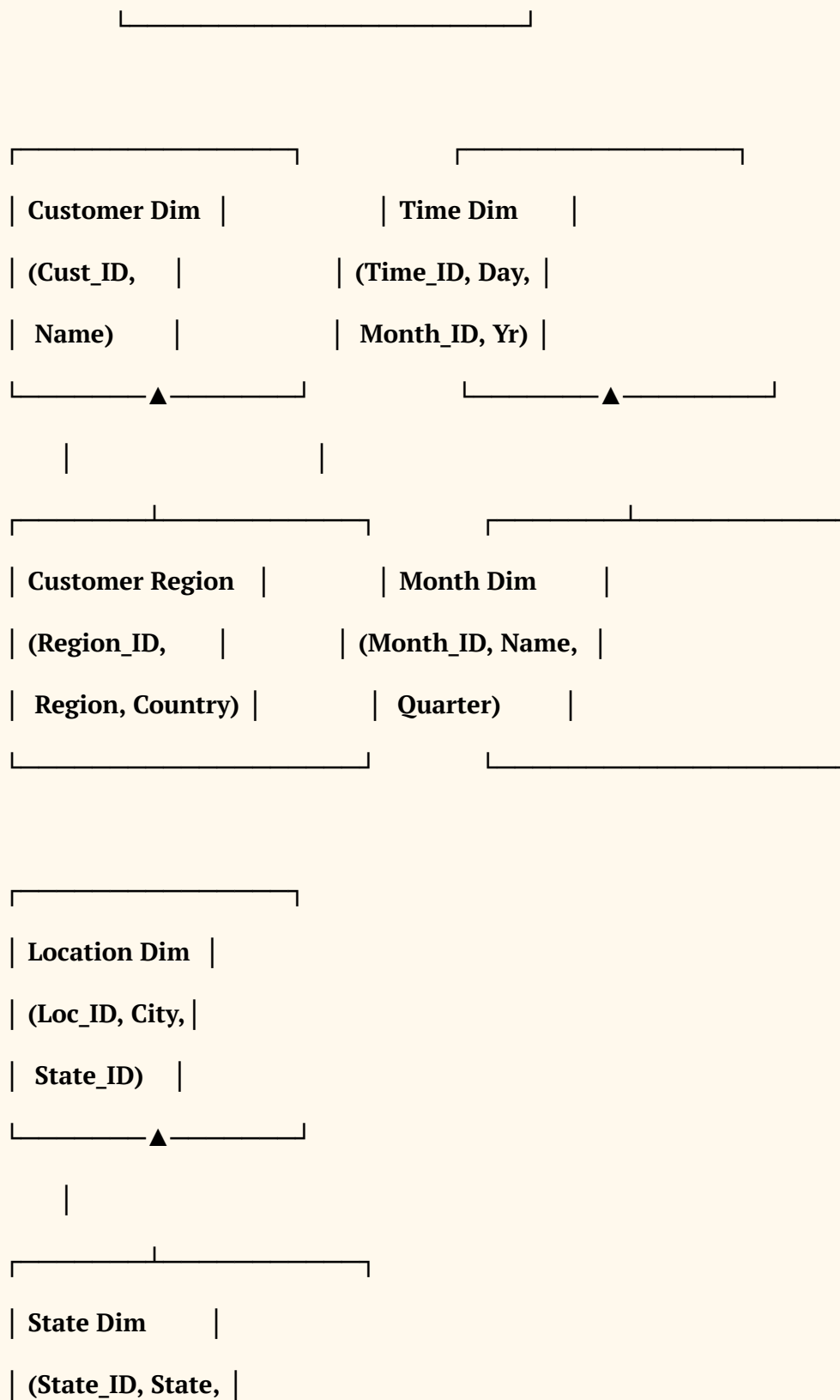
2. Star Schema

Structure

- In Star Schema, the Fact Table is at the center.
- It is surrounded by Denormalized Dimension Tables (directly connected).

Star Schema Diagram (Text Representation)





Country)

Sales Fact Table (Prod_ID, Cust_ID, Time_ID, Loc_ID, Quantity_sold, Amount_sold)

Explanation

- **Fact Table:** Same as star schema.
- **Dimensions are normalized:**
 - *Product Dim* split into Product and Product Category.
 - *Customer Dim* split into Customer and Customer Region.
 - *Time Dim* split into Time and Month.
 - *Location Dim* split into Location and State.

✓ **Advantages of Snowflake:** Less data redundancy, better for complex hierarchies.

✗ **Disadvantages:** More complex queries (needs multiple joins), slower query performance than Star Schema.

4. Final Answer (Concise for Exams)

- **Star Schema:** Central fact table surrounded by denormalized dimensions (easy, fast).
- **Snowflake Schema:** Fact table with normalized dimensions (less redundancy, more complex).
- For given case (Products, Customers, Time, Locations → Fact: Sales-Items with Quantity-sold & Amount-sold), the schemas are as drawn above.

(c) Define Noisy data while doing data preprocessing. Delete the noise with Binning smoothing techniques for the following details using partition in Bins (Equalfrequency) : 4, 2, 6, 10, 8, 16, 12, 24, 22, 14, 26 stored price details (in dollars).

Ans :

1. Definition of Noisy Data

- Noisy Data = data that contains random errors, outliers, or irrelevant variations.
- In data preprocessing, noise must be reduced or smoothed to improve data quality before applying mining techniques.

 **Example:** Price values with sudden large jumps or random errors are considered noise.

2. Given Dataset

Price details (in dollars):

4, 2, 6, 10, 8, 16, 12, 24, 22, 14, 26

3. Step 1 – Sort the Data

First, sort values in ascending order for binning:

2, 4, 6, 8, 10, 12, 14, 16, 22, 24, 26

4. Step 2 – Partition into Bins (Equal-frequency)

We have 11 values. Let's divide into 3 bins (each $\approx$ equal frequency).

- Bin size = $11 \div 3 \approx 3$ or 4 elements per bin.

So:

- Bin 1: 2, 4, 6, 8
- Bin 2: 10, 12, 14, 16
- Bin 3: 22, 24, 26

5. Step 3 – Apply Smoothing by Binning Techniques

(i) Smoothing by Bin Mean

Replace each value by the mean of its bin.

- Bin 1 mean = $(2+4+6+8)/4 = 5$
- Bin 2 mean = $(10+12+14+16)/4 = 13$
- Bin 3 mean = $(22+24+26)/3 = 24$

Smoothed data:

5, 5, 5, 5, 13, 13, 13, 13, 24, 24, 24

(ii) Smoothing by Bin Median

Replace each value by the median of its bin.

- Bin 1 median = $(4+6)/2 = 5$
- Bin 2 median = $(12+14)/2 = 13$
- Bin 3 median = 24

Smoothed data:

5, 5, 5, 5, 13, 13, 13, 13, 24, 24, 24

(Same result here since bins are symmetric.)

(iii) Smoothing by Bin Boundaries

Replace each value by the closest boundary (min or max) of the bin.

- Bin 1 boundaries = [2, 8] → values closer to 2 or 8
 - 2→2, 4→2, 6→8, 8→8
- Bin 2 boundaries = [10, 16]
 - 10→10, 12→10, 14→16, 16→16
- Bin 3 boundaries = [22, 26]
 - 22→22, 24→22, 26→26

Smoothed data:

2, 2, 8, 8, 10, 10, 16, 16, 22, 22, 26

6. Final Answer (Concise for Exams)

- Noisy data are data values containing random errors or outliers.
- Using Equal-frequency binning on given prices:

Sorted data: 2, 4, 6, 8 | 10, 12, 14, 16 | 22, 24, 26

- Bin Mean smoothing: 5, 5, 5, 5, 13, 13, 13, 13, 24, 24, 24
- Bin Median smoothing: 5, 5, 5, 5, 13, 13, 13, 13, 24, 24, 24
- Bin Boundary smoothing: 2, 2, 8, 8, 10, 10, 16, 16, 22, 22, 26

(d) Define Clustering in Data Mining. Write and explain k-means clustering algorithm. List its advantages and disadvantages.

Ans :

1. Definition of Clustering

- Clustering is an unsupervised learning technique in data mining that groups a set of data objects into clusters (groups) such that:
 - Objects in the same cluster are more similar to each other.
 - Objects in different clusters are dissimilar.

✓ It is widely used in pattern recognition, customer segmentation, image processing, and market research.

2. K-means Clustering Algorithm

- K-means is the most popular partition-based clustering algorithm.
- It partitions n data objects into k clusters, where k is predefined.
- Each cluster is represented by the mean (centroid) of its objects.

Algorithm Steps

1. **Input:** Choose the number of clusters k .
2. **Initialize:** Randomly select k data points as initial centroids.
3. **Assign:** Assign each data point to the nearest centroid (using distance measure, e.g., Euclidean distance).

4. **Update:** Recalculate the centroids as the mean of all points in that cluster.
5. **Repeat:** Steps 3–4 until centroids do not change (convergence) or a maximum number of iterations is reached.

Pseudocode

K-Means Algorithm:

1. Choose k (number of clusters).
2. Select k random points as initial centroids.
3. Repeat until convergence:
 - a. Assign each data point to the cluster with nearest centroid.
 - b. Update centroids by computing mean of cluster points.

Example (Simplified)

Suppose we have 6 data points on a 2D plane:

(2,3), (3,3), (6,7), (8,9), (5,8), (7,6)

- Choose $k = 2$.
- Randomly initialize centroids: $C1 = (2,3)$, $C2 = (8,9)$.
- Assign points → cluster around nearest centroid.
- Recalculate new centroids.
- Repeat until stable clusters are formed.

✓ Result → Two clusters formed: one around (2,3),(3,3) and the other around (6,7),(8,9),(5,8),(7,6).

3. Advantages of K-means

1. Simple & easy to implement.
2. Works well on large datasets.
3. Computationally efficient (time complexity: $O(n \cdot k \cdot t)$ where $t = \text{iterations}$).
4. Produces tighter clusters compared to hierarchical methods.

4. Disadvantages of K-means

1. Requires predefined k (number of clusters must be known).
2. Sensitive to initial centroids → may lead to different results.
3. Struggles with non-spherical clusters or clusters of varying density.
4. Sensitive to outliers (a single outlier can distort the mean).
5. May get stuck in local minima (not guaranteed to find global optimum).

✓ Final Exam-ready Answer (Summary)

- **Clustering:** Groups similar data into clusters.
- **K-means algorithm:** Iteratively partitions data into k clusters by minimizing the distance to centroids.
- **Advantages:** Simple, scalable, efficient.
- **Disadvantages:** Requires k in advance, sensitive to outliers, may converge to local optima.

2.

(a) What is Web-Mining ? List various webmining tasks. Also, discuss the following types of web-mining :

(i) Web content mining

(ii) Web usage mining

Ans :

1. Definition of Web Mining

- Web Mining is the process of applying data mining techniques to extract useful patterns, knowledge, and insights from web data.
- Web data includes web content (text, images, videos), web structure (links), and web usage (logs, clickstreams).

✓ It helps in information retrieval, personalization, business intelligence, digital marketing, and recommendation systems.

2. Web Mining Tasks

Some important tasks of web mining are:

1. **Resource Discovery** → Finding web documents, services, and information.
2. **Information Extraction** → Extracting structured information from unstructured web content.
3. **Generalization** → Finding patterns across multiple sites or web pages.
4. **Analysis** → Understanding user behavior, market trends, and site performance.
5. **Personalization** → Delivering customized content (e.g., Amazon recommendations).

3. Types of Web Mining

(i) Web Content Mining

- **Definition:** Process of extracting useful information from the content of web pages (text, images, audio, video, metadata).
- **Techniques Used:**
 - Natural Language Processing (NLP)
 - Information Retrieval (IR)
 - Machine Learning (text classification, clustering, sentiment analysis)
- **Example:**
 - Google search engine indexing pages based on keywords.
 - Sentiment analysis of Twitter posts about a product.

✓ **Use-case:** E-commerce sites mining product reviews to understand customer satisfaction.

(ii) Web Usage Mining

- **Definition:** Process of extracting useful knowledge from web logs, clickstream data, and user navigation patterns.
- **Techniques Used:**
 - Log file analysis
 - Association rule mining
 - Clustering of user sessions
- **Example:**
 - Amazon recommending "Customers who bought X also bought Y".

- Netflix suggesting movies based on past viewing behavior.

✓ Use-case: Optimizing website structure based on popular navigation paths.

✓ Final Exam-ready Answer (Summary)

- Web Mining: Application of data mining techniques to extract insights from web content, structure, and usage.
- Tasks: Resource discovery, information extraction, generalization, analysis, personalization.
- Types:
 1. Web Content Mining → Extracts data from web page content (text, multimedia). *Example: Google search, sentiment analysis.*
 2. Web Usage Mining → Extracts user behavior from logs/clickstreams. *Example: Amazon recommendations, website optimization.*

(b) With the help of an example, explain rulebased classification.

Ans :

1. Definition

- Rule-based classification is a classification technique that uses a set of IF-THEN rules to classify data into different classes.
- Each rule has two parts:
 - Condition (IF part / antecedent) → test on attribute values
 - Class label (THEN part / consequent) → predicted class

✓ Example Rule format:

IF (Age > 30 AND Income = High) THEN Class = "Yes"

2. Working Process

1. Rule Generation → Rules are extracted from training data using algorithms (like Decision Trees, RIPPER, or Apriori-based methods).
2. Rule Pruning → Remove unnecessary or conflicting rules.

3. **Classification** → A new data tuple is classified by testing it against the rules in sequence.

3. Example

Dataset (simplified)

Age	Student	Credit-rating	Buys Computer
<30	Yes	Fair	Yes
<30	No	Excellent	No
31-40	Yes	Fair	Yes
>40	No	Excellent	Yes
>40	Yes	Fair	No

Generated Rules

1. IF (Age = <30 AND Student = Yes) → Buys Computer = Yes
2. IF (Age = <30 AND Student = No) → Buys Computer = No
3. IF (Age = >40 AND Credit-rating = Excellent) → Buys Computer = Yes
4. ELSE → Buys Computer = No

Classification Example

- A new customer: (*Age = 25, Student = Yes, Credit = Fair*)
- Matches Rule 1 → Classified as Yes (Buys Computer).

4. Advantages

- Simple and intuitive ("IF-THEN" rules are easy to understand).
- Rules are interpretable and human-readable.
- Can handle both categorical and numerical data.

5. Disadvantages

- Rule generation can be computationally expensive.
- May generate a large number of rules (overfitting).
- Handling conflicts (when multiple rules match) requires strategies (e.g., highest confidence rule wins).

✓ Final Exam-ready Answer (Summary)

- Rule-based classification uses IF–THEN rules for predicting classes.
- Example rule: *IF (Age <30 AND Student = Yes) THEN Buys Computer = Yes.*
- It is easy to interpret but may suffer from overfitting and rule conflicts.

(c) What are the various steps involved in building a classification model ? Explain with the help of an example.

Ans :

Steps Involved in Building a Classification Model

Building a classification model involves several systematic steps:

1. Data Collection

- Gather relevant data from multiple sources (databases, logs, files).
- Data should represent the problem domain well.

✓ *Example:* Collect customer data (Age, Income, Education, Marital Status, Purchase History).

2. Data Preprocessing

- Clean and prepare the data for analysis.
- Handle missing values, noisy data, normalization, encoding categorical attributes.

✓ *Example:* Replace missing values of "Income" with the mean, convert "Marital Status" into numerical form.

3. Splitting the Dataset

- Divide the dataset into:
 - Training set → to build the model.
 - Test set → to evaluate accuracy.
- Common split: 70% training, 30% testing.

✓ *Example:* Out of 1000 customer records, use 700 for training and 300 for testing.

4. Model Building (Training Phase)

- Apply a classification algorithm (e.g., Decision Tree, Naïve Bayes, KNN, Rule-based, Neural Networks).
- The model learns the relationship between input features and class labels.

✓ *Example:* Use a Decision Tree to classify customers as "*Buy = Yes*" or "*Buy = No*" based on features like Age, Income, and Education.

5. Model Testing (Evaluation Phase)

- Apply the model on the test dataset.
- Measure accuracy using performance metrics such as:
 - Confusion Matrix
 - Accuracy, Precision, Recall, F1-score

✓ *Example:* The Decision Tree predicts correctly for 270 out of 300 test cases → Accuracy = 90%.

6. Model Deployment

- The model is deployed in a real-world system for classifying new unseen data.

✓ *Example:* An e-commerce site uses the trained model to predict if a new visitor is likely to purchase a product.

7. Model Maintenance & Updating

- Monitor model performance over time.

- Retrain or fine-tune with new data if performance decreases.

Illustrative Example (Simplified)

Dataset (Training):

Age	Income	Student	Buys Computer
<30	High	No	No
<30	High	Yes	Yes
31-40	Medium	Yes	Yes
>40	Low	No	No
>40	Medium	Yes	Yes

Decision Tree Constructed:

IF (Age < 30 AND Student = Yes) → Buys Computer = Yes

IF (Age < 30 AND Student = No) → Buys Computer = No

IF (Age > 40 AND Income = Medium) → Buys Computer = Yes

ELSE → Buys Computer = No

✓ Usage:

- New customer: (Age = 25, Income = High, Student = Yes)
- Rule matches → Classified as Yes (Buys Computer).

✓ Final Exam-ready Answer (Summary)

The steps in building a classification model are:

1. Data collection
2. Data preprocessing
3. Splitting dataset
4. Model training
5. Model testing/evaluation
6. Deployment
7. Maintenance

Example: A Decision Tree model can classify customers as "Buys Computer: Yes/No" based on attributes like Age, Income, and Student status.

Qus 3.

(a) With the help of an example, explain Market Basket Analysis.

Ans :

1. Definition

- Market Basket Analysis is a data mining technique used to discover association rules between items purchased together in transactions.
- It helps identify frequent itemsets and meaningful relationships.
- The output is usually in the form of IF-THEN association rules.

 **Example rule:**

IF {Milk, Bread} → THEN {Butter}

Meaning: Customers who buy *Milk* and *Bread* are also likely to buy *Butter*.

2. Process of Market Basket Analysis

1. Collect transaction data (e.g., receipts, invoices, POS data).
2. Find frequent itemsets using algorithms like Apriori or FP-Growth.
3. Generate association rules with measures like:
 - Support → How frequently items occur together.
 - Confidence → How often rule holds true.
 - Lift → Strength of association beyond random chance.

3. Example

Transaction Dataset (simplified):

Transaction ID	Items Bought
T1	Milk, Bread, Butter
T2	Milk, Bread
T3	Milk, Diaper, Beer, Bread
T4	Bread, Butter
T5	Milk, Diaper, Bread, Butter

Step 1: Frequent Itemsets Found

- {Milk, Bread} → appears in 4/5 transactions (Support = 80%).
- {Bread, Butter} → appears in 3/5 transactions (Support = 60%).

Step 2: Association Rules

- Rule 1: {Milk, Bread} → {Butter}
 - Support = 60%
 - Confidence = 75% (3 out of 4 times when Milk & Bread are bought, Butter is also bought).
- Rule 2: {Diaper} → {Beer}
 - Support = 40%
 - Confidence = 66%

4. Applications of Market Basket Analysis

- Retail: Product placement (putting related products together in stores).
- E-commerce: Recommendation engines (*"Customers who bought X also bought Y"*).

- Cross-selling: Suggesting complementary items.
- Inventory Management: Stocking items that are often bought together.

✓ Final Exam-ready Answer (Summary)

- Market Basket Analysis finds associations among items frequently bought together using association rules.
- Example: *IF {Milk, Bread} → THEN {Butter}*.
- It uses support, confidence, and lift to measure rule strength.
- Applications include retail product placement, e-commerce recommendations, and cross-selling.

(b) Write and explain Apriori algorithm used to identify the most frequently occurring elements and meaningful associations in any dataset.

Ans :

1. Definition

- Apriori algorithm is a classical data mining algorithm used to find frequent itemsets and generate association rules in transactional datasets.
- It works on the principle:
 👉 *If an itemset is frequent, then all of its subsets must also be frequent (Apriori property).*

2. Key Terms

- **Itemset:** A set of items (e.g., {Milk, Bread}).
- **Support:** Frequency of an itemset in transactions.

$$\text{Support}(X) = \frac{\text{Transactions containing } X}{\text{Total Transactions}}$$
- **Confidence:** Likelihood that item Y is purchased when item X is purchased.

$$\text{Confidence}(X \rightarrow Y) = \frac{\text{Support}(X \cup Y)}{\text{Support}(X)}$$
- **Lift:** Strength of rule compared to random chance.

$$\text{Lift}(X \rightarrow Y) = \frac{\text{Confidence}(X \rightarrow Y)}{\text{Support}(Y)}$$

3. Steps of Apriori Algorithm

1. Set minimum thresholds (min-support, min-confidence).
2. Generate Candidate Itemsets of length 1 (C1).
3. Prune itemsets that do not meet min-support → get Frequent Itemsets L1.
4. Generate Candidate Itemsets of length k from L(k-1).
5. Repeat pruning until no more frequent itemsets exist.
6. Generate Association Rules from frequent itemsets that meet min-confidence.

4. Example

Dataset (Transactions):

TID	Items Bought
T1	Milk, Bread, Butter
T2	Milk, Bread
T3	Milk, Diaper, Beer, Bread
T4	Bread, Butter
T5	Milk, Diaper, Bread, Butter

Step 1: Set thresholds

- min-support = 60% (≥ 3 transactions)
- min-confidence = 70%

Step 2: Generate Frequent Itemsets

- L1: {Milk} (4), {Bread} (5), {Butter} (3), {Diaper} (2), {Beer} (1)
→ Only {Milk}, {Bread}, {Butter} survive (support ≥ 3).

- L2: {Milk, Bread} (4), {Milk, Butter} (2), {Bread, Butter} (3)
→ Keep {Milk, Bread}, {Bread, Butter}.
- L3: {Milk, Bread, Butter} (2) → support < 3, discard.

Step 3: Generate Association Rules

- Rule 1: {Milk} → {Bread}
 - Confidence = $4/4 = 100\%$ ✓
- Rule 2: {Bread} → {Milk}
 - Confidence = $4/5 = 80\%$ ✓
- Rule 3: {Milk, Bread} → {Butter}
 - Confidence = $2/4 = 50\%$ ✗ (below threshold).

5. Applications of Apriori Algorithm

- Retail & E-commerce → Market Basket Analysis (Amazon recommendations).
- Healthcare → Finding co-occurrence of symptoms.
- Telecom → Identifying service usage patterns.
- Fraud Detection → Detecting unusual combinations of activities.

✓ Final Exam-ready Answer (Summary)

- Apriori algorithm finds frequent itemsets and association rules using the Apriori property.
- Steps: Generate candidate itemsets → prune → repeat until no more frequent itemsets → generate rules.
- Example: From supermarket data, rule {Milk} → {Bread} with 100% confidence.
- Applications: Retail, healthcare, telecom, fraud detection.

(c) List and discuss any two popular data mining tools.

Ans :

Two Popular Data Mining Tools

1. RapidMiner

- **Description:** An open-source data science platform used for data mining, machine learning, and predictive analytics.
- **Features:**
 - Provides a drag-and-drop interface (no need for complex coding).
 - Supports data preprocessing, modeling, evaluation, and deployment.
 - Integrates with Python, R, and Weka.
 - Has built-in operators for clustering, classification, and association rule mining.
- **Use Case Example:**
 - A bank can use RapidMiner to analyze customer transactions and predict loan defaults.

2. Weka (Waikato Environment for Knowledge Analysis)

- **Description:** A collection of machine learning algorithms written in Java, mainly for research and education.
- **Features:**
 - Supports classification, regression, clustering, association rules, and visualization.
 - Easy-to-use GUI plus command-line and Java API.
 - Works well with small to medium datasets.
 - Integrates with scripting languages like Python.
- **Use Case Example:**
 - A university research project can use Weka to classify student performance based on attendance, assignments, and test results.

✓ Final Exam-ready Answer (Summary)

- RapidMiner and Weka are two popular data mining tools.
- RapidMiner is widely used in industry for large-scale predictive analytics, while Weka is popular in research and education for experimenting with machine learning algorithms.

Qus 4.

(a) Discuss ETL and its need. Explain in detail, all the steps involved in ETL with the help of a suitable diagram.

Ans :

ETL (Extract, Transform, Load)

1. Definition

- ETL stands for Extract, Transform, Load.
- It is the data integration process used in data warehousing to:
 1. Extract data from multiple heterogeneous sources.
 2. Transform the data into a clean, consistent format
 3. Load the transformed data into a data warehouse for reporting & analysis.

2. Need for ETL

- Data is stored in different sources (databases, flat files, ERP systems, APIs).
- Raw data is inconsistent, noisy, incomplete → not suitable for analysis.
- ETL ensures:
 - Data consistency and quality.
 - Integration from multiple systems.
 - Historical storage in a warehouse.
 - Support for OLAP & BI tools for decision-making.

 **Example:** An e-commerce company extracts data from website logs, payment gateways, and CRM; transforms it into a unified format; and loads it into a data warehouse for customer behavior analysis.

3. Steps in ETL Process

(i) Extract

- Collect data from multiple heterogeneous sources:
 - Relational DBMS (MySQL, Oracle).
 - Flat files (CSV, JSON, XML).
 - Cloud systems (AWS, Salesforce).
- Can be full extraction (all data) or incremental extraction (only changes).

✓ *Example:* Extract customer info from CRM, sales data from POS, clickstream logs from website.

(ii) Transform

- Convert raw data into a clean, standardized format.
- Operations include:
 - Data Cleaning → handle missing/noisy data.
 - Data Integration → resolve conflicts between sources.
 - Data Transformation → normalization, encoding, aggregation, deriving new attributes.
- Ensures consistency, accuracy, usability.

✓ *Example:* Convert "M/F" into "Male/Female", standardize currency (₹ → \$), calculate "Total Sales = Quantity × Price".

(iii) Load

- Store the transformed data into the data warehouse.
- Two methods:
 - Full Load → all records loaded (initial load).
 - Incremental Load → only new/updated records added (ongoing).
- Can be loaded into Star Schema, Snowflake Schema, or Fact & Dimension tables.

✓ *Example:* Load customer dimension table and sales fact table in the warehouse.

4. ETL Process Diagram

Source Systems

(DBs, Files, APIs, ERP)

|

v

[Extract]

|

v

[Transform]

(Cleaning, Integration,
Standardization, Aggregation)

|

v

[Load]

|

v

Data Warehouse

(Star Schema / Snowflake Schema)

|

v

BI Tools, Reports, OLAP

5. Summary

- ETL (Extract, Transform, Load) is the backbone of data warehousing.
- It is needed to integrate heterogeneous, inconsistent, noisy data into a clean, consistent warehouse.
- Steps: Extract → Transform → Load.
- Supports business intelligence, OLAP, and better decision-making.

(b) List and explain any three key challenges of Data Warehouse.

Ans :

Key Challenges of Data Warehouse

A Data Warehouse (DW) integrates data from multiple sources to support decision-making. However, building and maintaining a DW faces many challenges.

Here are three key challenges:

1. Data Integration from Heterogeneous Sources

- **Problem:** Data comes from multiple sources (databases, ERP, flat files, cloud, web logs) in different formats, structures, and standards.
- **Challenge:**
 - Ensuring consistency (e.g., “Male/Female” vs “M/F”).
 - Resolving naming conflicts (e.g., “Cust_ID” vs “Customer_No”).
 - Handling different data types and encodings.
- **Impact:** If not handled properly, reports become inaccurate or misleading.

2. Data Quality and Cleaning

- **Problem:** Raw data often contains missing values, duplicates, noisy or inconsistent data.
- **Challenge:**
 - Removing errors (e.g., negative sales amount, impossible dates like 31-Feb).
 - Handling missing values (fill, ignore, or estimate).
 - Standardizing formats (e.g., DD/MM/YYYY vs MM/DD/YYYY).
- **Impact:** Poor data quality → poor decisions

3. Performance and Scalability

- **Problem:** Data warehouses deal with huge volumes of data (terabytes–petabytes).
- **Challenge:**
 - Ensuring fast query performance while handling large datasets.
 - Scaling infrastructure as data grows.
 - Balancing ETL load times with real-time analytics requirements.
- **Impact:** If queries are slow, BI tools and OLAP reports become less useful.

✓ Summary (Exam-ready)

Three key challenges of a Data Warehouse are:

1. **Data Integration** – combining heterogeneous data sources into one consistent warehouse.

2. **Data Quality** – cleaning missing, noisy, or inconsistent data.
3. **Performance & Scalability** – ensuring fast queries and scalability with growing data.

(c) With reference to Alex Gorelik, explain the following additional data lake stages :

(i) Data Puddle

(ii) Data Pond

(iii) Data Lake

(iv) Data Ocean

Ans :

Additional Data Lake Stages (as per Alex Gorelik)

(i) Data Puddle

- **Definition:** A very small, isolated dataset created for a specific individual or team's task.
- **Characteristics:**
 - Usually unmanaged, created quickly.
 - Stored in Excel files, small databases, or local systems.
 - No governance, metadata, or integration.
- **Example:** A sales analyst downloading customer leads into a spreadsheet for quick analysis.

(ii) Data Pond

- **Definition:** A departmental or business-unit-level data repository.
- **Characteristics:**
 - Supports one department (e.g., HR, Finance, Marketing).
 - Better structured than a puddle, but still siloed.
 - Limited sharing with other units.
- **Example:** Finance department building its own database of payroll, expenses, and budgets.

(iii) Data Lake

- **Definition:** A centralized enterprise repository that stores raw, semi-structured, and structured data.
- **Characteristics:**
 - Integrates data from multiple departments and sources.
 - Uses big data technologies (Hadoop, Spark, cloud storage).
 - Includes governance, metadata, and access control.
- **Example:** A retail company storing POS data, CRM records, clickstream logs, and IoT sensor data in a single big data platform.

(iv) Data Ocean

- **Definition:** The largest scope of data integration, extending beyond one enterprise to include multiple organizations, industries, or even global ecosystems.
- **Characteristics:**
 - Federation of multiple data lakes across companies, governments, or industries.
 - Enables large-scale collaboration, research, and AI.
- **Example:** A global healthcare ecosystem where hospitals, research labs, and governments share patient records, genomic data, and clinical trials to fight diseases.

Summary Table

Stage	Scope / Owner	Example
Data Puddle	Individual / Team	Analyst's Excel sheet
Data Pond	Department / Unit	Finance dept. payroll DB
Data Lake	Enterprise-wide	Company-wide big data platform

Data Ocean	Multi-org / Global	Worldwide healthcare data sharing
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✓ **Final Exam-Ready Answer:**

According to Alex Gorelik, the stages of data growth are:

1. Data Puddle – isolated, small datasets for individual use.
2. Data Pond – departmental repositories with limited scope.
3. Data Lake – centralized enterprise repository with governance.
4. Data Ocean – cross-organization, global data ecosystems.

5. Write short notes on the following :

(a) Aggregate fact table and derived dimensional tables

(b) Data swamp

(c) Data Preprocessing stages

(d) Agglomerative approach of Hierarchical method

Ans :

Additional Data Lake Stages (as per Alex Gorelik)

(i) Data Puddle

- **Definition:** A very small, localized dataset created for a specific task or team.
- **Features:**
 - Usually stored in spreadsheets, local databases, or small cloud buckets.
 - Created without long-term governance or enterprise standards.
- **Example:** A marketing analyst downloads customer data from CRM into Excel for a campaign.
- **Pros/Cons:** Quick and flexible ✓ but isolated and inconsistent ✗.

(ii) Data Pond

- **Definition:** A departmental-level collection of data, slightly larger than a puddle, used by a specific business unit.
- **Features:**

- Serves a single department (e.g., Finance, HR, Sales).
- More structured than a puddle but still lacks enterprise-wide integration.
- Example: Finance team creates a database of expense and payroll data to analyze departmental costs.
- Pros/Cons: Better collaboration within a team ✓ but silos across departments ✗.

(iii) Data Lake

- Definition: A centralized repository that stores raw, semi-structured, and structured data from multiple sources across the organization.
- Features:
 - Supports big data technologies (Hadoop, Spark).
 - Enables analytics, ML, and data science.
 - Follows governance, metadata, and access control.
- Example: An enterprise collects IoT data, CRM data, logs, and social media feeds in a single data lake for advanced analytics.
- Pros/Cons: Scalable & flexible ✓ but requires strong governance ✗.

(iv) Data Ocean

- Definition: The largest-scale data repository, extending beyond an enterprise to include multiple organizations, industries, or even global ecosystems.
- Features:
 - Interconnected data lakes across companies, industries, or governments.
 - Data sharing across ecosystems.
- Example: A global healthcare consortium pools patient, genomic, and clinical trial data across hospitals worldwide.
- Pros/Cons: Extremely powerful for research & innovation ✓ but has huge privacy, security, and governance challenges ✗.

Summary Table

Stage	Scope	Example
Data Puddle	Individual / Local use	Analyst's Excel file

Data Pond	Department / Business Unit	Finance dept. expense DB
Data Lake	Enterprise-wide	Central big data repository
Data Ocean	Cross-organization / Global	Global healthcare data sharing

 Final Exam-Ready Answer:

According to Alex Gorelik, data evolves in four stages:

1. **Data Puddle** – small, local datasets for individual use.
2. **Data Pond** – departmental data repositories.
3. **Data Lake** – centralized enterprise-wide data repository.
4. **Data Ocean** – global, cross-organization interconnected data ecosystems.

June, 2024

1.

(a) With the help of a diagram, explain the real-time data warehouse architecture. Also, discuss its trade-off.

Ans :

Real-Time Data Warehouse (RTDW) Architecture

✓ Definition

A real-time data warehouse is a system where data is continuously captured, transformed, and loaded into the warehouse so that users can access up-to-date, current information for decision-making. Unlike traditional DWs (batch ETL overnight), RTDW provides instant or near real-time analytics.

Architecture Components (with Diagram)

Source Systems (ERP, CRM, IoT, Logs, Social Media)

|

▼

Real-Time Data Capture

(Change Data Capture, Streaming APIs)

|

▼

Real-Time ETL Layer

(Data Cleaning, Transformation,

Stream Processing: Kafka, Spark)

|



Real-Time Data Warehouse Storage

(Columnar DB, Cloud DW: Snowflake, BigQuery)



Metadata & Governance Layer



BI / Analytics / Dashboards / Alerts

(OLAP, Reporting, ML, Visualization Tools)

Step-by-Step Flow

1. **Source Systems** – Operational systems like ERP, CRM, IoT sensors, web logs.
2. **Real-Time Data Capture** – Uses Change Data Capture (CDC) or streaming APIs to detect changes immediately.
3. **Real-Time ETL Layer** – Data is cleaned, transformed, and enriched in-stream (using tools like Apache Kafka, Spark Streaming, Flink).
4. **Data Warehouse Storage** – Data is stored in a real-time capable warehouse (Snowflake, Google BigQuery, Amazon Redshift, etc.).
5. **Metadata & Governance** – Ensures data consistency, lineage, and security.
6. **Analytics & Access Layer** – Business users access dashboards, OLAP cubes, and real-time alerts.

Trade-offs in Real-Time Data Warehouse

✅ Advantages (Pros):

- **Timely decisions:** Supports fraud detection, stock trading, supply chain monitoring.
- **Competitive edge:** Enables real-time personalization in e-commerce.

- **Operational efficiency:** Detects issues instantly (e.g., failed transactions).

✗ Challenges (Cons / Trade-offs):

1. **High Cost & Complexity** – Needs advanced streaming platforms, high-performance DBs, and skilled staff.
2. **Performance Overhead** – Continuous ETL can slow down systems if not optimized.
3. **Data Quality Risks** – Harder to ensure accurate cleaning/transformation in real time compared to batch.
4. **Scalability & Maintenance** – Requires strong infrastructure to handle streaming at scale.

✓ Final Exam-Ready Answer

A Real-Time Data Warehouse continuously integrates operational data into a warehouse for instant analysis.

Its architecture includes Source Systems → Real-Time Capture → Streaming ETL → Warehouse Storage → Metadata → BI Tools.

Trade-off: It provides up-to-date insights and fast decisions, but comes at the cost of higher complexity, infrastructure needs, and data quality challenges.

(b) Describe fact constellation scheme. How is it different from snow-flake scheme ? With the help of an example explain fact constellation schema. Mention its advantages and limitations

Ans :

Fact Constellation Schema

✓ Definition

- A Fact Constellation Schema (also called Galaxy Schema) is a multidimensional schema that contains multiple fact tables sharing dimension tables.
- It is a collection of stars → hence the term "constellation."
- Used when business processes are complex and require analysis across multiple fact tables.

Difference between Snowflake Schema and Fact Constellation Schema

Aspect	Snowflake Schema	Fact Constellation Schema
Fact Tables	Usually one fact table with normalized dimensions.	Multiple fact tables sharing common dimensions.
Complexity	Moderate (normalized star schema).	High (many stars interlinked).
Usage	When dimensions are large and require normalization.	When multiple business processes need to be modeled together.
Representation	Looks like a snowflake shape due to normalized dimension tables.	Looks like a galaxy of stars (fact constellation).

Example of Fact Constellation Schema

Use Case: Retail Company

- The company wants to analyze Sales and Shipping together.

Fact Tables:

1. Sales Fact Table

- Measures: *Quantity Sold, Amount Sold*
- Dimensions: *Customer, Product, Time, Location*

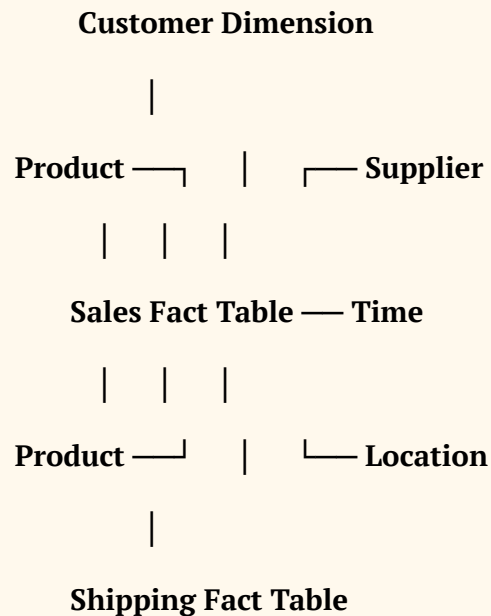
2. Shipping Fact Table

- Measures: *Shipping Cost, Delivery Time*
- Dimensions: *Customer, Product, Time, Location, Supplier*

Shared Dimensions:

- Customer, Product, Time, Location are shared dimensions across both fact tables.

📌 Diagram (Simplified)



This looks like a galaxy of stars = Fact Constellation.

Advantages of Fact Constellation Schema

1. **Handles complex business processes (multiple fact tables).**
2. **Data sharing → Common dimensions reduce redundancy.**
3. **Flexibility → Supports enterprise-wide analytics across processes (sales, shipping, inventory).**

Limitations

1. **Complex design → Harder to build and maintain.**
2. **Query performance → More joins required, so queries can be slower.**
3. **Higher storage & governance needs compared to star schema.**

✅ Final Exam-Ready Answer

A Fact Constellation Schema (or Galaxy Schema) uses multiple fact tables that share dimension tables. It differs from the Snowflake Schema, which has a single fact table with normalized dimensions.

Example: In retail, *Sales* and *Shipping* fact tables may share dimensions like *Customer*, *Product*, *Time*, and *Location*.

Advantages: Supports complex analytics and reduces redundancy.

Limitations: Design and query processing are more complex.

(c) Write and explain K-NN classification algorithm in Data-Mining. Discuss its advantages and disadvantages.

Ans :

K-Nearest Neighbor (K-NN) Classification Algorithm

✓ Definition

- K-NN (K-Nearest Neighbors) is a lazy learning, non-parametric classification algorithm.
- It classifies a new data point based on the majority class of its K nearest neighbors in the feature space.
- The “closeness” is measured using a distance metric (e.g., Euclidean distance, Manhattan distance).

Steps in K-NN Algorithm

1. **Choose K** → The number of nearest neighbors to consider (e.g., K = 3, K = 5).
2. **Compute Distances** → For a new data point, calculate the distance from all training points.
3. **Select Nearest Neighbors** → Pick the K closest points.
4. **Voting / Majority Rule** → Assign the class that is most common among those K neighbors.
5. **Classify the new data point accordingly.**

Example

Suppose we have a dataset with two classes:

- Class A (Blue) $\rightarrow (1,2), (2,3), (3,3)$
- Class B (Red) $\rightarrow (6,6), (7,7), (8,6)$

👉 A new point (4,4) comes.

- Compute distances to all points.
- If $K = 3$, the nearest neighbors are (3,3), (6,6), (2,3).
- Among them: 2 are Class A, 1 is Class B.
- ✅ The new point (4,4) is classified as Class A.

Advantages of K-NN

1. Simple and Intuitive – Easy to understand and implement.
2. No Training Phase – Lazy learning; works instantly with data.
3. Versatile – Can be used for both classification and regression.
4. Works well on small datasets with clear class boundaries.

Disadvantages of K-NN

1. Computationally Expensive – Needs distance calculation for every prediction.
2. Sensitive to Noise & Irrelevant Features – Can misclassify outliers.
3. Choice of K matters – Too small $K \rightarrow$ overfitting, too large $K \rightarrow$ underfitting.
4. Not good for high-dimensional data (suffers from *curse of dimensionality*).

✅ Final Exam-Ready Answer

The K-NN algorithm classifies new data points by finding the K nearest neighbors and assigning the majority class among them.

It is simple and effective but becomes slow and less accurate with large or high-dimensional datasets.

(d) Discuss vector space modeling for representing text documents. With reference to this modeling, explain TF-IDF and Inverse Document Frequency (IDF).

Ans :

Vector Space Model (VSM) for Representing Text Documents

✓ Definition

- The Vector Space Model (VSM) is a mathematical model used to represent text documents as vectors of terms (words).
- Each document is transformed into a vector in multi-dimensional space, where each dimension corresponds to a unique term from the vocabulary.
- This allows us to compute similarity between documents using measures like cosine similarity.

How it Works

1. Build a vocabulary of all terms across documents.
2. Represent each document as a vector:
 - Example: $D1 = (t1, t2, t3, ..., tn)$
3. Each value in the vector represents the weight of the term in the document (based on TF-IDF or other weighting methods).

TF-IDF Weighting in Vector Space Model

1. Term Frequency (TF)

- Measures how frequently a term occurs in a document.
- Formula:

$$TF(t, d) = \frac{\text{Number of times term } t \text{ appears in document } d}{\text{Total number of terms in document } d}$$

👉 Example:

In document $D1 = \text{"Data mining is mining useful data"}$,

- Term "mining" occurs 2 times,
- Total terms = 5,
- So, $TF(\text{mining}, D1) = 2/5 = 0.4$.

2. Inverse Document Frequency (IDF)

- Measures how important a term is across all documents.
- If a word occurs in many documents, it is less useful (e.g., "the", "is").
- Formula:

$$IDF(t) = \log\left(\frac{N}{df(t)}\right)$$

Where:

- N = total number of documents
- $df(t)$ = number of documents containing term t

👉 Example:

Suppose we have 10 documents, and the term "mining" occurs in 2 of them.

$$IDF(\text{mining}) = \log\left(\frac{10}{2}\right) = \log(5) \approx 0.699$$

3. TF-IDF (Combined Weighting)

- Combines TF and IDF to assign a weight to each term.
 - Formula:
- $$TF\text{-}IDF(t, d) = TF(t, d) \times IDF(t)$$

👉 Example (from above):

- $TF(\text{mining}, D1) = 0.4$
 - $IDF(\text{mining}) = 0.699$
 - So,
- $$TF\text{-}IDF(\text{mining}, D1) = 0.4 \times 0.699 \approx 0.28$$

✅ Summary (Exam-Ready Answer)

- In the Vector Space Model, documents are represented as vectors of weighted terms.
- TF measures the frequency of a term in a document.
- IDF reduces the weight of common terms and increases the weight of rare but important terms.

- **TF-IDF = TF × IDF** gives a balanced weight, allowing better document comparison in text mining and information retrieval.

Qus 2.

(a) Discuss various factors to be considered to improve the performance of ETL.

Ans :

Factors to Improve the Performance of ETL

ETL (Extract, Transform, Load) is the backbone of a data warehouse. Since ETL processes deal with large volumes of data, optimizing performance is critical. The following factors should be considered:

1. Extraction Optimization

- **Incremental Extraction** → Extract only the new or changed data, instead of full data reload.
- **Parallel Extraction** → Extract from multiple sources simultaneously.
- **Filtering at Source** → Apply filters at the source database instead of bringing unnecessary data into staging.

2. Transformation Optimization

- **Push-Down Processing** → Perform transformations at the database/source system instead of the ETL tool (reduces data movement).
- **Efficient Use of Joins** → Use indexed joins and avoid unnecessary Cartesian products.
- **Data Cleansing in Staging** → Clean data in the staging area to reduce workload later.
- **Avoid Row-by-Row Processing** → Prefer set-based operations instead of looping through rows.

3. Loading Optimization

- **Bulk Loading** → Use database-specific bulk load utilities (e.g., Oracle SQL*Loader, PostgreSQL COPY).
- **Partitioning** → Partition large fact tables and load in parallel.

- **Disable Indexes & Constraints Temporarily** → Disable indexes before bulk load and rebuild after loading to speed up inserts.
- **Batch Commit** → Commit after batches (not row-by-row) to reduce over

4. Infrastructure & Resource Management

- **Parallelism & Multi-threading** → Run ETL jobs in parallel pipelines.
- **Memory Tuning** → Allocate sufficient memory for transformations and caching.
- **Network Optimization** → Reduce data transfer bottlenecks between ETL server and data warehouse.
- **Scheduling & Load Balancing** → Run ETL jobs during off-peak hours, distribute workload evenly.

5. Metadata and Monitoring

- **Metadata-Driven ETL** → Automate and optimize based on metadata (data lineage, source info).
- **Monitoring & Logging** → Track job run-times, identify bottlenecks, and tune queries.

✓ Summary (Exam-Ready Answer)

To improve ETL performance, organizations should focus on:

- **Efficient Extraction** (incremental, parallel, filtering at source)
- **Optimized Transformation** (push-down processing, set-based operations)
- **Optimized Loading** (bulk load, partitioning, disabling indexes)
- **Infrastructure Tuning** (parallelism, memory, scheduling)
- **Monitoring with Metadata** (continuous performance checks)

These practices ensure faster ETL pipelines, reduced resource usage, and timely data availability in the data warehouse.

(b) Define a Data Mart. How is it different from a centralized data warehouse ? Discuss the structure of Data Mart with the help of an example.

Ans :

✓ Definition

A Data Mart is a subset of a data warehouse, focused on a specific business function, department, or subject area such as sales, finance, marketing, or HR.

- It contains summarized and relevant data rather than the complete enterprise data.
- Data marts are used by specific groups of users for faster decision-making.

Difference Between Data Mart and Centralized Data Warehouse

Aspect	Data Mart	Centralized Data Warehouse
Scope	Department-specific / Subject-specific	Organization-wide
Data Volume	Smaller (subset of data)	Very large (all enterprise data)
Purpose	Focused analysis for a team or business unit	Comprehensive analysis across the enterprise
Implementation Time	Quick (weeks to months)	Long (months to years)
Complexity	Simple and easy to maintain	Complex with large schema
Cost	Lower	Higher
Users	Departmental managers, analysts	Executives, data scientists, enterprise users

Structure of a Data Mart (with Example)

Example: Sales Data Mart

Suppose a retail company wants to analyze sales performance.

- **Fact Table** → Contains numerical measures (e.g., *Quantity Sold*, *Amount Sold*).
- **Dimension Tables** → Provide context for the facts.

Sales Data Mart Schema (Star Schema)

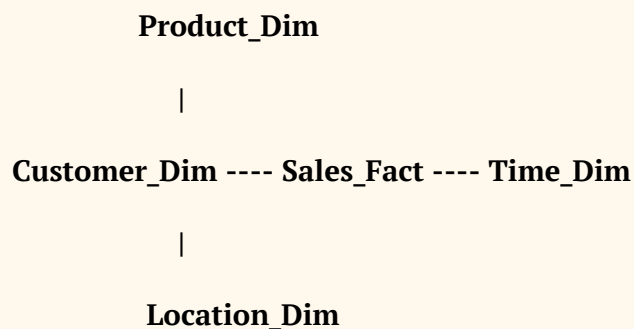
Fact Table:

- **Sales_Fact**
 - Quantity_Sold
 - Amount_Sold
 - Product_ID (FK)
 - Customer_ID (FK)
 - Time_ID (FK)
 - Location_ID (FK)

Dimension Tables:

- **Product_Dim** → Product_ID, Product_Name, Category, Brand
- **Customer_Dim** → Customer_ID, Name, Age, Segment
- **Time_Dim** → Time_ID, Day, Month, Year
- **Location_Dim** → Location_ID, City, State, Region

Diagram (Star Schema for Data Mart)



👉 Here, the Sales_Fact table stores measurable data, while dimensions add descriptive details for analysis.

✅ **Summary (Exam-Ready Answer)**

- A Data Mart is a department-focused subset of a data warehouse designed for faster, subject-specific analysis.
- It differs from a centralized data warehouse in scope, size, cost, and implementation time.
- Example: A Sales Data Mart can have a fact table (sales data) linked with dimension tables (products, customers, time, locations), often represented in a star schema structure.

(c) Discuss the following OLAP data cube operations :

(i) Roll-up

(ii) Drill-down.

Ans :

OLAP Data Cube Operations

OLAP (Online Analytical Processing) supports multidimensional analysis of data using data cubes.

Two of the most important operations are Roll-up and Drill-down.

(i) Roll-up (Aggregation)

✅ **Definition:**

- The Roll-up operation summarizes data at a higher level of abstraction by climbing up the concept hierarchy or reducing dimensions.
- It is also called aggregation.

✅ **How it works:**

1. Aggregates measures by combining data (e.g., sum, average).

2. Moves from detailed data → summarized data.

✓ Example:

- Suppose we have a Sales data cube with dimensions: *Time* (*Day* → *Month* → *Year*), *Location* (*City* → *State* → *Country*), and *Product*.
- Roll-up example:
 - From daily sales → monthly sales.
 - Or, from City-level sales → Country-level sales.

(ii) Drill-down (De-aggregation)

✓ Definition:

- The Drill-down operation is the reverse of roll-up.
- It provides a finer level of detail by descending the concept hierarchy or adding more dimensions.
- Also called disaggregation.

✓ How it works:

1. Explores data at a more granular level.
2. Moves from summarized data → detailed data.

✓ Example:

- From Yearly sales → Quarterly sales → Monthly sales → Daily sales.
- Or, from Country sales → State sales → City sales.

 Diagram (Conceptual View)

Drill-down (more detail)

Country → State → City

Roll-up (more summary)

✓ Summary (Exam-Ready Answer)

- **Roll-up:** Aggregates data at a higher level (e.g., City → Country, Day → Year).
- **Drill-down:** Moves into finer levels of detail (e.g., Year → Month → Day).
- Together, they enable multi-level data analysis in OLAP systems, helping decision-makers view both summarized and detailed perspectives.

Qus 3.

(a) Discuss the following Data Integration issues :

(i) Schema Integration and object modeling

(ii) Redundancy

(iii) Detection and Resolution of Data Value Conflicts.

Ans :

Data Integration Issues

Data Integration is the process of combining data from multiple heterogeneous sources (databases, files, web services, etc.) into a unified view for analysis in a data warehouse. While doing so, several challenges/issues arise:

(i) Schema Integration and Object Modeling

- **Problem:** Different data sources often use different schemas, names, data types, and structures for the same concept.
- **Challenge:**
 - Attribute naming conflicts → e.g., "*Cust_ID*" in one source vs "*Customer_No*" in another.
 - Structural differences → One source stores *Address* as a single field, another splits it into *Street*, *City*, *State*.
 - Data type mismatches → *Date* as "YYYY-MM-DD" in one source vs "DD/MM/YYYY" in another.
- **Solution:**
 - Use schema matching and object modeling techniques.
 - Create a global schema to map different local schemas into one unified representation.

(ii) Redundancy

- **Problem:** When combining multiple sources, the same data may be repeated → leads to duplication and inconsistency.
- **Example:**
 - Customer “Ritesh Tiwari” may appear in multiple databases (sales, marketing, support).
 - One entry has address as “Howrah, WB” and another as “Howrah, West Bengal”.
- **Impact:**
 - Increases storage cost.
 - Leads to confusion during analysis (which record is correct?).
- **Solution:**
 - Apply duplicate detection and record linkage techniques.
 - Use data cleansing tools to merge duplicates.

(iii) Detection and Resolution of Data Value Conflicts

- **Problem:** Even if schema and structure are integrated, the actual values of data may conflict.
- **Examples:**
 - Salary of an employee is ₹50,000 in one system and ₹55,000 in another.
 - Date of birth stored differently (1999-02-01 vs 01-Feb-1999).
- **Challenge:** Which value is correct? How to resolve inconsistencies?
- **Solution:**
 - Define data precedence rules (decide which source is more reliable).
 - Use data transformation (standard formats, units, currency conversions).
 - Apply business rules to reconcile conflicts (e.g., latest update overrides old one).

✓ Summary (Exam-Ready Answer)

Data Integration faces the following issues:

1. **Schema Integration & Object Modeling** → Different attribute names, structures, and types must be unified under a global schema.
2. **Redundancy** → Duplicate records across sources must be detected and cleaned.

3. **Data Value Conflicts** → Inconsistent values across sources must be resolved using precedence rules, transformations, or business rules.

These challenges must be handled properly to ensure high-quality, consistent, and reliable data in the data warehouse.

(b) Write and explain Decision Tree Classifier algorithm with the help of an example.

Ans :

Decision Tree Classifier Algorithm

✓ Definition

A Decision Tree is a supervised learning classification model that represents decisions and their possible outcomes as a tree-like structure.

- **Internal nodes** → represent an attribute (feature).
- **Branches** → represent a test outcome.
- **Leaves** → represent a class label (final decision).

It is widely used in classification and prediction because of its simplicity and interpretability.

✓ Decision Tree Algorithm (Steps)

1. **Start with Training Dataset** → with attributes (features) and class labels.
2. **Select the best attribute for the root node using a splitting criterion:**
 - Information Gain (ID3 algorithm)
 - Gain Ratio (C4.5 algorithm)
 - Gini Index (CART algorithm)
3. **Create a node for the selected attribute.**
4. **Split the dataset into subsets based on attribute values.**
5. **Repeat steps 2–4 recursively for each subset until:**
 - All tuples belong to the same class, OR
 - No more attributes are left, OR
 - A stopping condition is reached.

6. Assign class labels to leaf nodes.

✓ Example: Playing Tennis

Suppose we want to decide whether to Play Tennis based on weather conditions.

Dataset attributes:

- Outlook (Sunny, Overcast, Rainy)
- Temperature (Hot, Mild, Cool)
- Humidity (High, Normal)
- Wind (Weak, Strong)
- Class Label: PlayTennis = {Yes, No}

Step 1: Choose Best Attribute

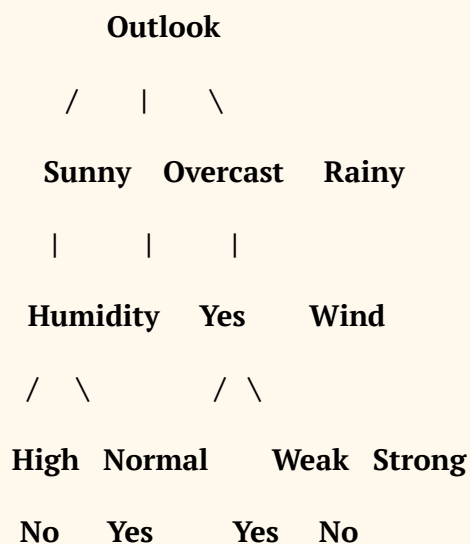
Using Information Gain, the attribute Outlook has the highest gain.

So, Outlook becomes the root node.

Step 2: Split Data

- If Outlook = Overcast → PlayTennis = Yes
- If Outlook = Sunny → further split on Humidity
- If Outlook = Rainy → further split on Wind

Step 3: Build the Tree



✓ Final Decision Tree Classifier

- If Outlook = Overcast $\rightarrow$ PlayTennis = Yes
- If Outlook = Sunny and Humidity = High $\rightarrow$ PlayTennis = No
- If Outlook = Sunny and Humidity = Normal $\rightarrow$ PlayTennis = Yes
- If Outlook = Rainy and Wind = Weak $\rightarrow$ PlayTennis = Yes
- If Outlook = Rainy and Wind = Strong $\rightarrow$ PlayTennis = No

✓ Strengths of Decision Trees

- Easy to understand and visualize.
- Handles both numerical and categorical data.
- Requires little data preprocessing.

✓ Weaknesses of Decision Trees

- Can easily overfit the data.
- Small changes in data may result in a completely different tree.
- Less efficient for very high-dimensional data.

📌 Exam-Ready Answer (Summary)

The Decision Tree Classifier is a supervised learning algorithm that recursively splits data based on the best attribute (using Information Gain, Gain Ratio, or Gini Index) until class labels are assigned at leaves.

Example: In the Play Tennis dataset, the root node is Outlook, and the final tree makes classification decisions like “If Outlook = Overcast, then Play = Yes.”

4.

(a) Discuss Linear Discriminant Analysis (LDA) and Principal Component Analysis (PCA) feature extraction techniques.

Ans :

Feature Extraction Techniques: LDA vs PCA

Feature extraction techniques reduce the dimensionality of data while retaining important information. Two widely used techniques are:

1. Linear Discriminant Analysis (LDA)

✓ Definition

- LDA is a supervised feature extraction technique.
- It finds a linear combination of features that best separates two or more classes.
- Unlike PCA, it uses class labels.

✓ Working

1. Compute the within-class scatter (variation within each class).
2. Compute the between-class scatter (variation between class means).
3. Find the projection (linear transformation) that maximizes the ratio of between-class scatter to within-class scatter.
4. Project the data onto this lower-dimensional space for classification.

✓ Example

- Suppose we classify students as Pass or Fail based on two features: Study Hours and IQ.
- LDA finds the best line (axis) that maximizes separation between Pass and Fail classes.
- New data points can then be classified by projecting onto this line.

✓ Key Use

- Used in classification problems (e.g., face recognition, medical diagnosis).

2. Principal Component Analysis (PCA)

✓ Definition

- PCA is an unsupervised feature extraction technique.

- It reduces dimensionality by finding new axes (principal components) that capture the maximum variance in the data.
- Does not use class labels.

✓ Working

1. Standardize the dataset.
2. Compute the covariance matrix.
3. Calculate eigenvalues and eigenvectors.
4. Select the top k eigenvectors (principal components) corresponding to the largest eigenvalues.
5. Project the data onto this lower-dimensional subspace.

✓ Example

- Suppose we have 3 features for employees: *Age*, *Salary*, *Years of Experience*.
- PCA may find that *Salary* and *Experience* are highly correlated.
- It combines them into a single principal component, reducing redundancy while preserving variance.

✓ Key Use

- Used in data compression, noise reduction, visualization, and preprocessing for machine learning.

LDA vs PCA (Comparison Table)

Aspect	LDA (Linear Discriminant Analysis)	PCA (Principal Component Analysis)
Type	Supervised (uses class labels)	Unsupervised (no class labels)
Goal	Maximize class separation	Maximize variance
Projection	Based on class scatter matrices	Based on covariance matrix

Best for	Classification tasks	Dimensionality reduction, preprocessing
Example	Face recognition (class separation)	Image compression (variance capture)

✓ Exam-Ready Summary

- **LDA** → Supervised method that reduces dimensions by maximizing separation between classes. Best for **classification tasks**.
- **PCA** → Unsupervised method that reduces dimensions by maximizing data variance. Best for **compression and preprocessing**.

(b) Explain Association Rule Mining (ARM). Also, explain the two phases involved in ARM process with the help of an example.

Ans :

Association Rule Mining (ARM)

✓ Definition

- **Association Rule Mining (ARM)** is a **data mining technique** used to find **hidden patterns, correlations, or associations** among items in large datasets.
- It is most famously used in **Market Basket Analysis**.

Example:

In a supermarket dataset:

- **{Milk, Bread} → {Butter}**
means *if a customer buys Milk and Bread, then they are also likely to buy Butter.*

✓ Form of Association Rule

An association rule is written as:

$$X \rightarrow Y \text{ } \Rightarrow Y$$

Where:

- XX = antecedent (left-hand side, purchased items)
- YY = consequent (right-hand side, likely purchased item)

✓ Two Phases of ARM Process

Phase 1: Frequent Itemset Generation

- Find all **itemsets** (combinations of items) that occur **frequently** in the dataset.
- Use a threshold called **minimum support**.
- **Support (X)** = Proportion of transactions that contain itemset X.

Example:

If out of 1,000 transactions, 200 contain {Milk, Bread}, then:

$$\text{Support}(\text{Milk, Bread}) = \frac{200}{1000} = 20\%$$

Phase 2: Rule Generation

- From the frequent itemsets, generate association rules that satisfy a minimum **confidence**.
- **Confidence (X → Y)** = Likelihood that a transaction containing X also contains Y.

$$\text{Confidence}(X \rightarrow Y) = \frac{\text{Support}(X \cup Y)}{\text{Support}(X)} \quad \text{Confidence}(X \Rightarrow Y) = \frac{\text{Support}(X \cup Y)}{\text{Support}(X)}$$

Example:

- If 200 transactions contain {Milk, Bread}, and 150 of those also contain {Butter}, then:

$$\text{Confidence}(\text{Milk, Bread} \rightarrow \text{Butter}) = \frac{150}{200} = 75\% \quad \text{Confidence}(\text{Milk, Bread} \Rightarrow \text{Butter}) = \frac{150}{200} = 75\%$$

This means 75% of the time, people who buy Milk & Bread also buy Butter.

✓ Example (Market Basket Analysis)

Suppose we have a dataset of transactions:

TID	Items Bought
1	Milk, Bread, Butter
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Bread, Butter
4	Bread, Milk, Diaper, Beer
5	Milk, Bread, Butter, Diaper
	• Frequent Itemset: {Milk, Bread, Butter} • Rule: {Milk, Bread} → {Butter} • Support: $3/5 = 60\%$ • Confidence: $3/3 = 100\%$

✓ Summary (Exam-Ready Answer)

- **Association Rule Mining (ARM)** discovers patterns of co-occurrence in large datasets.
- It works in **two phases**:
 1. **Frequent Itemset Generation** (items meeting minimum support).
 2. **Rule Generation** (rules meeting minimum confidence).
- Example: In market basket analysis, a rule like {Milk, Bread} → {Butter} shows strong association, useful for **cross-selling, recommendations, and promotions**.

5. Write short notes on the following :

(a) **ELT and its need**

(b) **Metadata and Data Warehousing**

(c) **Rule based classification**

(d) Naive-payes Classifier

Ans :

(a) ELT and its Need

- **Definition:** ELT (Extract–Load–Transform) is a variation of ETL where data is **extracted** from sources, **loaded** into the data warehouse (or data lake) first, and then **transformed** within the target system itself.
- **Why Needed?**
 - Traditional ETL transforms data before loading, which may be slow for **big data**.
 - ELT leverages the **power of modern data warehouses** (e.g., Snowflake, BigQuery, Redshift) to perform transformations more efficiently.
- **Use Case:** When handling **large-scale raw data** in cloud-based warehouses.

✓ **Key Advantage:** Faster processing, scalability, and reduced dependency on external ETL servers.

(b) Metadata and Data Warehousing

- **Definition:** Metadata means “*data about data*”.
- In **data warehousing**, metadata defines:
 - **Technical Metadata** → data source, format, extraction methods.
 - **Business Metadata** → meaning of data (e.g., “Customer_ID” = unique identifier of a customer).
 - **Operational Metadata** → data refresh time, data lineage, error logs.
- **Importance in DW:**
 - Helps users understand **where the data came from** and how it is transformed.
 - Enables **query optimization** and efficient retrieval.
 - Ensures **consistency and trust** in data warehouse reports.

✓ **Example:** A metadata entry may describe that *Sales_Date* is stored in **YYYY-MM-DD** format and refreshed **daily at midnight**.

(c) Rule-Based Classification

- **Definition:** A classification method where a set of **IF-THEN rules** are used to predict the class of new records.
- **Example:**
 - Rule 1: *IF Age < 25 AND Income = High THEN Class = "Student"*
 - Rule 2: *IF Age > 40 AND Income = Low THEN Class = "Retired"*
- **How It Works:**
 - Rules are generated from training data.
 - When a new tuple comes, the system matches it with rules and assigns the class.
- **Advantages:** Easy to understand, interpretable.
- **Disadvantages:** Rule explosion (too many rules), handling conflicting rules can be tricky.

(d) Naïve Bayes Classifier

- **Definition:** A **probabilistic classifier** based on **Bayes' Theorem**, assuming independence among features.
- **Formula:**

$$P(C|X) = \frac{P(X|C) \cdot P(C)}{P(X)}$$

Where $P(C|X)$ is the probability of class C given features X .
- **How It Works:**
 - Compute prior probabilities of classes (from training data).
 - Compute likelihood for each feature given class.
 - Apply Bayes' rule and classify the data into the class with **highest posterior probability**.
- **Example:** Spam Email Classification:
 - Words like "free", "win", "offer" → high probability of Spam.
 - Words like "invoice", "meeting" → high probability of Not Spam.
- **Advantages:** Fast, works well with high-dimensional data (text mining).
- **Disadvantages:** Assumes feature independence, which may not always hold true.

✓ Summary (for exams)

- **ELT:** Extract → Load → Transform, needed for big data in cloud warehouses.
- **Metadata:** Data about data, essential in DW for consistency and trust.
- **Rule-Based Classification:** IF-THEN rules for classifying records.

- **Naïve Bayes:** Probabilistic classifier using Bayes' theorem, good for text mining and spam detection.